6 / DIGITAL CIRCUITS

Even in a large-scale digital system, such as a computer, or a data-processing, control, or digital-communication system, there are only a few basic operations which must be performed. These operations, to be sure, may be repeated very many times. The four circuits most commonly employed in such systems are known as the OR, AND, NOT, and FLIP-FLOP. These are called *logic* gates, or circuits, because they are used to implement Boolean algebraic equations (as we shall soon demonstrate). This algebra was invented by G. Boole in the middle of the nineteenth century as a system for the mathematical analysis of logic.

This chapter discusses in detail the first three basic logic circuits mentioned above. These basic gates are combined into FLIP-FLOPS and other digital-system building blocks in Chap. 17.

6-1 DIGITAL (BINARY) OPERATION OF A SYSTEM

A digital system functions in a binary manner. It employs devices which exist only in two possible states. A transistor is allowed to operate at cutoff or in saturation, but not in its active region. A node may be at a high voltage of, say, 4 ± 1 V or at a low voltage of, say, 0.2 ± 0.2 V, but no other values are allowed. Various designations are used for these two quantized states, and the most common are listed in Table 6-1. In logic, a statement is characterized as *true* or *false*, and this is the first binary classification listed in the table. A switch may be *closed* or *open*, which is the notation under 9, etc. Binary arithmetic and mathematical manipulation of switching or logic functions are best carried out with classification 3, which involves two symbols, 0 (zero) and 1 (one).

TABLE 6-1 Binary-state terminology

	1	2	3	4	5	6	7	8	9	10	11
One of the states.....	True	High	1	Up	Pulse	Excited	Off	Hot	Closed	North	Yes
The other state.....	False	Low	0	Down	No pulse	Non-excited	On	Cold	Open	South	No

The binary system of representing numbers will now be explained by making reference to the familiar *decimal system.* In the latter the base is 10 (ten), and ten numerals, 0, 1, 2, 3, . . . , 9, are required to express an arbitrary number. To write numbers larger than 9, we assign a meaning to the *position* of a numeral in an array of numerals. For example, the number 1,264 (one thousand two hundred sixty four) has the meaning

$$1{,}264 \equiv 1 \times 10^3 + 2 \times 10^2 + 6 \times 10^1 + 4 \times 10^0$$

Thus the individual digits in a number represent the coefficients in an expansion of the number in powers of 10. The digit which is farthest to the right is the coefficient of the zeroth power, the next is the coefficient of the first power, and so on.

In the *binary system* of representation the base is 2, and only the two numerals 0 and 1 are required to represent a number. The numerals 0 and 1 have the same meaning as in the decimal system, but a different interpretation is placed on the position occupied by a digit. In the binary system the individual digits represent the coefficients of powers of *two* rather than *ten* as in the decimal system. For example, the decimal number 19 is written in the binary representation as 10011 since

$$\begin{aligned} 10011 &\equiv 1 \times 2^4 + 0 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0 \\ &= \quad 16 \quad + \quad 0 \quad + \quad 0 \quad + \quad 2 \quad + \quad 1 \quad = 19 \end{aligned}$$

A short list of equivalent numbers in decimal and binary notation is given in Table 6-2.

A general method for converting from a decimal to a binary number is indicated in Table 6-3. The procedure is the following. Place the decimal number (in this illustration, 19) on the extreme right. Next divide by 2 and place the quotient (9) to the left and indicate the remainder (1) directly below it. Repeat this process (for the next column $9 \div 2 = 4$ and a remainder of 1) until a quotient of 0 is obtained. The array of 1's and 0's in the second row is the binary representation of the original decimal number. In this example, decimal 19 = 10011 binary.

A binary digit (a 1 or a 0) is called a *bit.* A group of bits having a significance is a *bite, word,* or *code.* For example, to represent the 10 numerals

TABLE 6-2 Equivalent numbers in decimal and binary notation

Decimal notation	*Binary notation*	*Decimal notation*	*Binary notation*
0	00000	11	01011
1	00001	12	01100
2	00010	13	01101
3	00011	14	01110
4	00100	15	01111
5	00101	16	10000
6	00110	17	10001
7	00111	18	10010
8	01000	19	10011
9	01001	20	10100
10	01010	21	10101

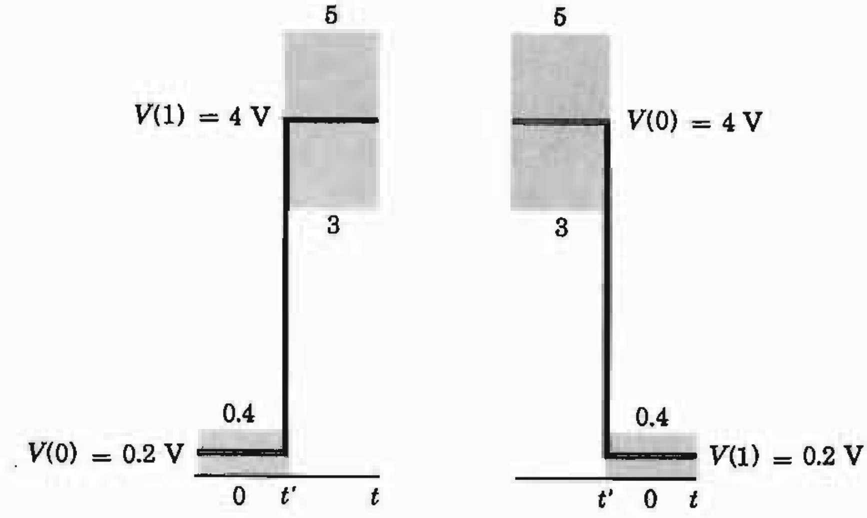
(0, 1, 2, . . . , 9) and the 26 letters of the English alphabet would require 36 different combinations of 1's and 0's. Since $2^5 < 36 < 2^6$, then a minimum of 6 bits per bite are required in order to accommodate all the alphanumeric characters. In this sense a bite is sometimes referred to as a *character* and a group of one or more characters as a *word.*

Logic Systems In a *dc*, or *level-logic*, system a bit is implemented as one of two voltage levels. If, as in Fig. 6-1*a*, the more positive voltage is the 1 level and the other is the 0 level, the system is said to employ dc *positive* logic. On the other hand, a dc *negative*-logic system, as in Fig. 6-1*b*, is one which designates the more negative voltage state of the bit as the 1 level and the more positive as the 0 level. It should be emphasized that the absolute values of the two voltages are of no significance in these definitions. In particular, the 0 state need not represent a zero voltage level (although in some systems it might).

The parameters of a physical device (for example, $V_{CE,\text{sat}}$ of a transistor) are not identical from sample to sample, and they also vary with temperature. Furthermore, ripple or voltage spikes may exist in the power supply or ground leads, and other sources of unwanted signals, called *noise*, may be present in the circuit. For these reasons the digital levels are not specified precisely, but as indicated by the shaded regions in Fig. 6-1, each state is defined by a voltage range about a designated level, such as 4 ± 1 V and 0.2 ± 0.2 V.

TABLE 6-3 Decimal-to-binary conversion

Divide by 2..........	0	1	2	4	9	19 decimal
Remainder...........	1	0	0	1	1	Binary



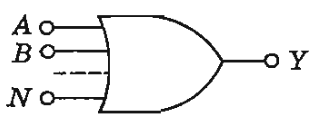
(a) (b)

Fig. 6-1 Illustrating the definitions of (a) positive and (b) negative logic. A transition from one state to the other occurs at $t = t'$.

In a *dynamic*, or *pulse-logic*, system a bit is recognized by the presence or absence of a pulse. A 1 signifies the existence of a positive pulse in a dynamic positive-logic system; a negative pulse denotes a 1 in a dynamic negative-logic system. In either system a 0 at a particular input (or output) at a given instant of time designates that no pulse is present at that particular moment.

6-2 THE OR GATE[1]

An OR gate has two or more inputs and a single output, and it operates in accordance with the following definition: *The output of an* OR *assumes the* 1 *state if one or more inputs assume the* 1 *state.* The n inputs to a logic circuit will be designated by $A, B, \ldots, N$ and the output by Y. It is to be understood that each of these symbols may assume one of two possible values, either 0 or 1. A standard symbol for the OR circuit is given in Fig. 6-2a, together with the Boolean expression for this gate. The equation is to be read "Y equals A *or* B *or* $\cdot\cdot\cdot$ *or* N." Instead of defining a logical operation in words, an alternative method is to give a *truth table* which contains a tabulation of all possible input values and their corresponding outputs. It



$Y = A + B + \cdots + N$

(a)

Input		Output
A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

(b)

Fig. 6-2 (a) The standard symbol for an OR gate and its Boolean expression; (b) the truth table for a two-input OR gate.

should be clear that the two-input truth table of Fig. 6-2*b* is equivalent to the above definition of the OR operation.

In a *diode-logic* (DL) system the logical gates are implemented by using diodes. A diode OR for negative logic is shown in Fig. 6-3. The generator source resistance is designated by R_s. We consider first the case where the supply voltage V_R has a value equal to the voltage $V(0)$ of the 0 state for dc logic.

If all inputs are in the 0 state, the voltage across each diode is $V(0) - V(0) = 0$. Since, in order for a diode to conduct, it must be forward-biased by at least the cutin voltage V_γ (Fig. 4-5), none of the diodes conducts. Hence the output voltage is $v_o = V(0)$, and Y is in the 0 state.

If now input A is changed to the 1 state, which for negative logic is at the potential $V(1)$, less positive than the 0 state, then $D1$ will conduct. The output becomes

$$v_o = V(0) - [V(0) - V(1) - V_\gamma]\frac{R}{R + R_s + R_f} \tag{6-1}$$

where R_f is the diode forward resistance. Usually R is chosen much larger than $R_s + R_f$. Under this restriction

$$v_o \approx V(1) + V_\gamma \tag{6-2}$$

Hence the output voltage exceeds the more negative level $V(1)$ by V_γ (approximately 0.2 V for germanium or 0.6 V for silicon). Furthermore, the step in output voltage is *smaller* by V_γ than the change in input voltage.

From now on, unless explicitly stated otherwise, we shall assume $R \gg R_s$ and ideal diodes with $R_f = 0$ and $V_\gamma = 0$. The output, for input A excited, is then $v_o = V(1)$, and the circuit has performed the following logic: if $A = 1$, $B = 0, \ldots, N = 0$, then $Y = 1$, which is consistent with the OR operation.

For the above excitation, the output is at $V(1)$, and each diode, except $D1$, is back-biased. Hence the presence of signal sources at $B, C, \ldots, N$ does not result in an additional load on generator A. Since the OR configuration minimizes the interaction of the sources on one another, this gate is sometimes referred to as a *buffer* circuit. Since it allows several independent sources to be applied at a given node, it is also called a (nonlinear) *mixing* gate.

If two or more inputs are in the 1 state, the diodes connected to these inputs conduct and all other diodes remain reverse-biased. The output is $V(1)$, and again the OR function is satisfied. If for any reason the level $V(1)$ is not identical for all inputs, *the most negative value of* $V(1)$ *(for negative logic) appears at the output,* and all diodes except one are nonconducting.

A positive-logic OR gate uses the same configuration as that in Fig. 6-3, except that all diodes must be reversed. *The output now is equal to the most positive level* $V(1)$ [or more precisely is smaller than the most positive value of $V(1)$ by V_γ]. If a dynamic logic system is under consideration, *the output-*

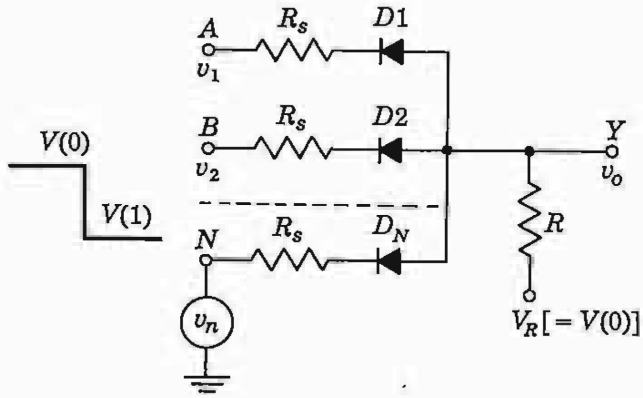


Fig. 6-3 A diode OR circuit for negative logic. [It is also possible to choose the supply voltage such that $V_R > V(0)$, but that arrangement has the disadvantage of drawing standby current when all inputs are in the 0 state.]

pulse magnitude is (approximately) *equal to the largest input pulse* (regardless of whether the system uses positive or negative logic).

A second mode of operation of the OR circuit of Fig. 6-3 is possible if V_R is set equal to a voltage more positive than $V(0)$ by at least V_γ. For this condition *all diodes conduct in the* 0 *state*, and $v_o \approx V(0)$ if $R \gg R_s + R_f$. If one or more inputs are excited, then the diode connected to the most negative $V(1)$ conducts, the output equals this value of $V(1)$, and all other diodes are back-biased. Clearly, the OR function has been satisfied.

A third mode of operation of the circuit of Fig. 6-3 results if we select $V_R < V(0)$. This arrangement has the disadvantage that the output will not respond until the input falls enough to overcome the initial reverse bias of the diodes.

Boolean Identities If it is remembered that A, B, and C can take on only the value 0 or 1, the following equations from Boolean algebra pertaining to the OR (+) operation are easily verified:

$$A + B + C = (A + B) + C = A + (B + C) \tag{6-3}$$

$$A + B = B + A \tag{6-4}$$

$$A + A = A \tag{6-5}$$

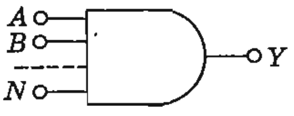
$$A + 1 = 1 \tag{6-6}$$

$$A + 0 = A \tag{6-7}$$

These equations may be justified by referring to the definition of the OR operation, to a truth table, or to the action of the OR circuits discussed above.

6-3 THE AND GATE[1]

An AND gate has two or more inputs and a single output, and it operates in accordance with the following definition: *The output of an* AND *assumes the* 1 *state if and only if all the inputs assume the* 1 *state.* A symbol for the AND



$Y = AB \cdots N$

(*a*)

Input		Output
A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

(*b*)

Fig. 6-4 (*a*) The standard symbol for an AND gate and its Boolean expression; the truth table for a two-input AND gate.

circuit is given in Fig. 6-4*a*, together with the Boolean expression for this gate. The equation is to be read "Y equals A *and* B *and* . . . *and* N." [Sometimes a dot (·) or a cross (×) is placed between symbols to indicate the AND operation.] It may be verified that the two-input truth table of Fig. 6-4*b* is consistent with the above definition of the AND operation.

A diode-logic (DL) configuration for a negative AND gate is given in Fig. 6-5*a*. To understand the operation of the circuit, assume initially that all source resistances R_s are zero and that the diodes are ideal. If *any* input is at the 0 level $V(0)$, the diode connected to this input conducts and the output is clamped at the voltage $V(0)$, or $Y = 0$. However, if *all* inputs are at the 1 level $V(1)$, then all diodes are reverse-biased and $v_o = V(1)$, or $Y = 1$. Clearly, the AND operation has been implemented. The AND gate is also called a *coincidence circuit.*

A positive-logic AND gate uses the same configuration as that in Fig. 6-5*a*, except that all diodes are reversed. This circuit is indicated in Fig. 6-5*b* and should be compared with Fig. 6-3. It is to be noted that the symbol $V(0)$ in Fig. 6-3 designates the same voltage as $V(1)$ in Fig. 6-5*b* because each represents the upper binary level. Similarly, $V(1)$ in Fig. 6-3 equals $V(0)$ in Fig. 6-5*b*, since both represent the lower binary level. Hence these two circuits are identical, and we conclude that *a negative* OR *gate is the same circuit as a positive* AND *gate.* This result is not restricted to diode logic, and by using Boolean algebra, we show in Sec. 6-8 that it is valid independently of the hardware used to implement the circuit.

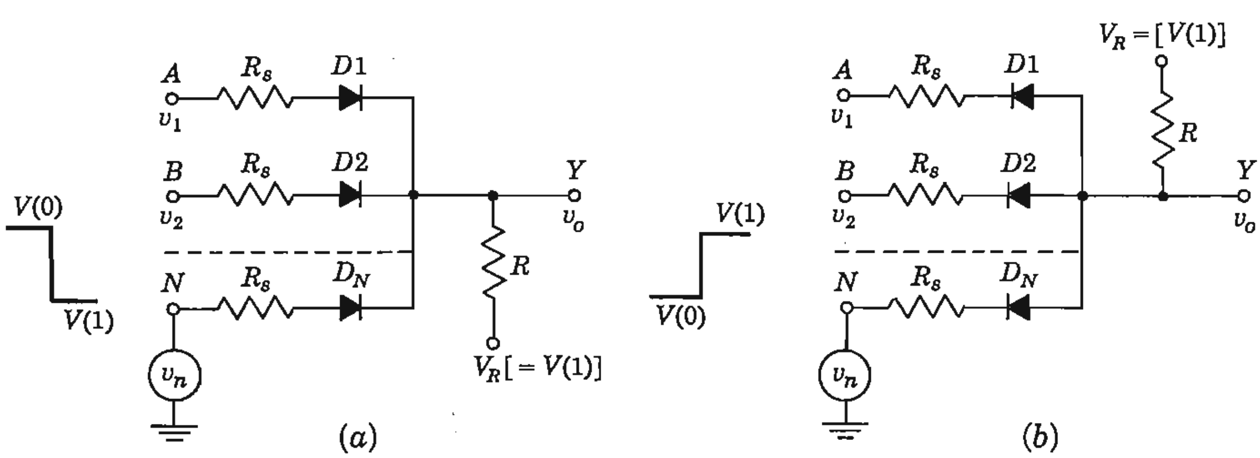


Fig. 6-5 A diode-logic AND circuit for (*a*) negative logic and (*b*) positive logic.

In Fig. 6-5*b* it is possible to choose V_R to be more positive than $V(1)$. If this condition is met, all diodes will conduct upon a coincidence (all inputs in the 1 state) and the output will be clamped to $V(1)$. The output impedance is low in this mode of operation, being equal to $(R_s + R_f)/n$ in parallel with R. On the other hand, if $V_R = V(1)$, then all diodes are cut off at a coincidence, and the output impedance is high (equal to R). If for any reason not all imputs have the same upper level $V(1)$, then the output of the positive AND gate of Fig. 6-5*b* will equal $V(1)_{\min}$, the *least* positive value of $V(1)$. Note that the diode connected to $V(1)_{\min}$ conducts, clamping the output to this minimum value of $V(1)$ and maintaining all other diodes in the reverse-biased condition. If, on the other hand, V_R is smaller than all inputs $V(1)$, then all diodes will be cut off upon coincidence and the output will rise to the voltage V_R. Similarly, if the inputs are pulses, then *the output pulse will have an amplitude equal to the smallest input amplitude* [provided that V_R is greater than $V(1)_{\min}$].

Boolean Identities Since A, B, and C can have only the value 0 or 1, the following expressions involving the AND operation may be verified:

$$ABC = (AB)C = A(BC) \tag{6-8}$$

$$AB = BA \tag{6-9}$$

$$AA = A \tag{6-10}$$

$$A1 = A \tag{6-11}$$

$$A0 = 0 \tag{6-12}$$

$$A(B + C) = AB + AC \tag{6-13}$$

These equations may be proved by reference to the definition of the AND operation, to a truth table, or to the behavior of the AND circuits discussed above. Also, by using Eqs. (6-11), (6-13), and (6-6), it can be shown that

$$A + AB = A \tag{6-14}$$

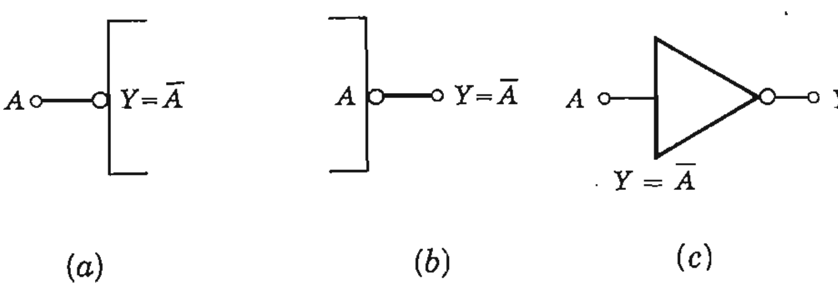
Similarly, if follows from Eqs. (6-13), (6-10), and (6-6) that

$$A + BC = (A + B)(A + C) \tag{6-15}$$

We shall have occasion to refer to the last two equations later.

6-4 THE NOT, OR INVERTER, CIRCUIT

The NOT circuit has a single input and a single output and performs the operation of *logic negation* in accordance with the following definition: *The output of a* NOT *circuit takes on the* 1 *state if and only if the input does* not *take on the* 1 *state.* The standard to indicate a *logic negation* is a small circle drawn at



Input	Output
A	Y
0	1
1	0

(d)

Fig. 6-6 Logic negation at (a) the input and (b) the output of a logic block; (c) a symbol often used for a NOT gate and the Boolean equation; (d) the truth table.

the point where a signal line joins a logic symbol. Negation at the input of a logic block is indicated in Fig. 6-6*a* and at the output in Fig. 6-6*b*. The symbol for a NOT gate and the Boolean expression for negation are given in Fig. 6-6*c*. The equation is to be read "Y equals NOT A" or "Y is the complement of A." [Sometimes a prime (′) is used instead of the bar (¯) to indicate the NOT operation.] The truth table is given in Fig. 6-6*d*.

A circuit which accomplishes a logic negation is called a NOT circuit, or, since it inverts the sense of the output with respect to the input, it is also known as an *inverter*. The output of an INVERTER is relatively more positive if and only if the input is relatively less positive. In a truly binary system only two levels $V(0)$ and $V(1)$ are recognized, and the output, as well as the input, of an inverter must operate between these two voltages. When the input is at $V(0)$, the output must be at $V(1)$, and vice versa. Ideally, then, a NOT circuit inverts a signal while preserving its shape and the binary levels between which the signal operates.

The transistor circuit of Fig. 6-7 implements an inverter for positive logic having a 0 state of $V(0) = V_{EE}$ and a 1 state of $V(1) = V_{CC}$. If the input is low, $v_i = V(0)$, then the parameters are chosen so that the Q is OFF, and hence $v_o = V_{CC} = V(1)$. On the other hand, if the input is high, $v_i = V(1)$, then the circuit parameters are picked so that Q is in saturation and then $v_o = V_{EE} = V(0)$, if we neglect the collector-to-emitter saturation voltage $V_{CE,\text{sat}}$. A detailed calculation of quiescent conditions is made in the following example.

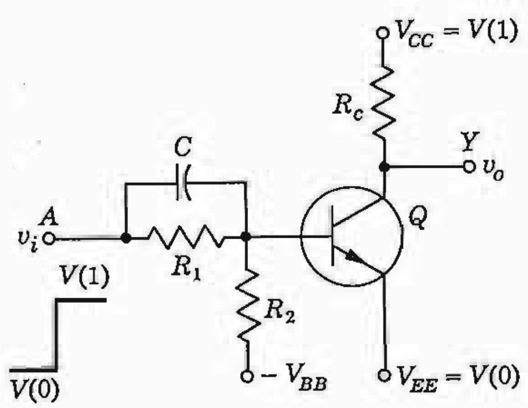


Fig. 6-7 An INVERTER for positive logic. A similar circuit using a *p-n-p* transistor is used for a negative-logic NOT circuit.

EXAMPLE If the silicon transistor in Fig. 6-8 has a minimum value of h_{FE} of 30, find the output levels for input levels of 0 and 12 V.

Solution For $v_i = V(0) = 0$ the open-circuited base voltage V_B is

$$V_B = -12 \times \frac{15}{100 + 15} = -1.56 \text{ V}$$

Since a bias of about 0 V is adequate to cut off a silicon emitter junction (Table 5-1, page 142), then Q is indeed cut off. Hence $v_o = 12$ V for $v_i = 0$.

For $v_i = V(1) = 12$ V let us verify the assumption that Q is in saturation. The minimum base current required for saturation is

$$(I_B)_{\min} = \frac{I_C}{h_{FE}}$$

It is usually sufficiently accurate to use the approximate values for the saturation junction voltages given in Table 5-1, which for silicon are $V_{BE,\text{sat}} = 0.8$ V and $V_{CE,\text{sat}} = 0.2$ V. With these values

$$I_C = \frac{12 - 0.2}{2.2} = 5.36 \text{ mA} \qquad (I_B)_{\min} = \frac{5.36}{30} = 0.18 \text{ mA}$$

$$I_1 = \frac{12 - 0.8}{15} = 0.75 \text{ mA} \qquad I_2 = \frac{0.8 - (-12)}{100} = 0.13 \text{ mA}$$

and

$$I_B = I_1 - I_2 = 0.75 - 0.13 = 0.62 \text{ mA}$$

Since this value exceeds $(I_B)_{\min}$, Q is indeed in saturation and the drop across the transistor is $V_{CE,\text{sat}}$. Hence $v_o = 0.2$ V for $v_i = 12$ V, and the circuit has performed the NOT operation.

If the input to the inverter is obtained from the output of a similar gate, the input levels are $V(0) = V_{CE,\text{sat}} = 0.2$ V and $V(1) = 12$ V. The corresponding output levels are 12 and 0.2 V, respectively.

The capacitor C across R_1 in Fig. 6-7 is added to improve the transient response of the inverter. This capacitor aids in the removal of the minority-carrier charge stored in the base when the signal changes abruptly between

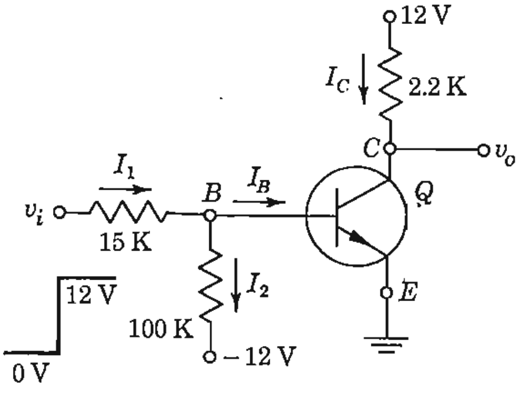


Fig. 6-8 An inverter calculation.

logic states. A discussion of this phenomenon, including rise time, fall time, and storage time, is given in the following section. The order of magnitude of C is 100 pF, but its exact value depends upon the transistor.

Transistor Limitations There are certain transistor characteristics as well as certain circuit features which must particularly be taken into account in designing transistor inverters.

1. *The back-bias emitter-junction voltage* V_{EB}. This voltage must not exceed the emitter-to-base breakdown voltage BV_{EBO} specified by the manufacturer. For the type 2N914, $BV_{EBO} = 5$ V, and for the 2N1304, $BV_{EBO} = 25$ V. However, for some (diffused-base) transistors, BV_{EBO} may be quite small (less than 1 V).

2. *The dc current gain* h_{FE}. Since h_{FE} decreases with decreasing temperature, the circuit must be designed so that at the lowest expected temperature the transistor will remain in saturation. The maximum value of R_1 is determined principally by this condition.

3. *The reverse collector saturation current* I_{CBO}. Since $|I_{CBO}|$ increases about 7 percent/°C (doubles every 10°C for either germanium or silicon), we cannot continue to neglect the effect of I_{CBO} at high temperatures. At cutoff the emitter current is zero and the base current is I_{CBO} (in a direction opposite to that indicated as I_B in Fig. 6-8). Let us calculate the value of I_{CBO} which just brings the transistor to the point of cutoff. If we assume, as in Table 5-1, that at cutoff, $V_{BE} = 0$ V, then $I_1 = 0$ and the drop across the 100-K resistor is

$$100\ I_{CBO} = 12\ \text{V} \qquad \text{or} \qquad I_{CBO} = 0.12\ \text{mA}$$

The ambient temperature at which $I_{CBO} = 0.12$ mA $= 120\ \mu$A is the maximum temperature at which the inverter will operate satisfactorily. A silicon transistor can be operated at temperatures in excess of 185°C.

Boolean Identities From the basic definition of the NOT, AND, and OR connectives we can verify the following Boolean identities:

$$\bar{\bar{A}} = A \tag{6-16}$$

$$\bar{A} + A = 1 \tag{6-17}$$

$$\bar{A}A = 0 \tag{6-18}$$

$$A + \bar{A}B = A + B \tag{6-19}$$

EXAMPLE Verify Eq. (6-19).

Solution Since $B + 1 = 1$ and $A1 = A$, then

$$A + \bar{A}B = A(B + 1) + \bar{A}B = AB + A + \bar{A}B = (A + \bar{A})B + A = B + A$$

where use is made of Eq. (6-17).

6-5 TRANSISTOR SWITCHING TIMES[2]

If a pulse is applied to an inverter, the transistor acts as a switch, since it operates from cutoff to saturation and then returns to cutoff. We now turn our attention to the behavior of the transistor inverter as it makes a transition from one state to the other. We consider the transistor circuit shown in Fig. 6-9*a*, driven by the pulse waveform shown in Fig. 6-9*b*. This waveform makes transitions between the voltage levels V_2 and V_1. At V_2 the transistor is at cutoff, and at V_1 the transistor is in saturation. The input waveform v_i is applied between base and emitter through a resistor R_s, which represents $R_1 \| R_2$ of Fig. 6-7 (assume that C is not present).

The response of the collector current i_C to the input waveform, together with its time relationship to that waveform, is shown in Fig. 6-9*c*. The current does not immediately respond to the input signal. Instead, there is a delay, and the time that elapses during this delay, together with the time required for the current to rise to 10 percent of its maximum (saturation) value $I_{CS} \approx V_{CC}/R_L$, is called the *delay time* t_d. The current waveform has a nonzero *rise time* t_r, which is the time required for the current to rise through the active region from 10 to 90 percent of I_{CS}. The total *turn-on* time t_{ON} is the sum of the delay and rise time, $t_{ON} \equiv t_d + t_r$. When the input signal returns to its initial state at $t = T$, the current again fails to respond immediately. The interval which elapses between the transition of the input waveform and the time when i_C has dropped to 90 percent of I_{CS} is called the *storage time* t_s. The storage interval is followed by the *fall time* t_f, which is the time required for i_C to fall from 90 to 10 percent of I_{CS}. The *turnoff*

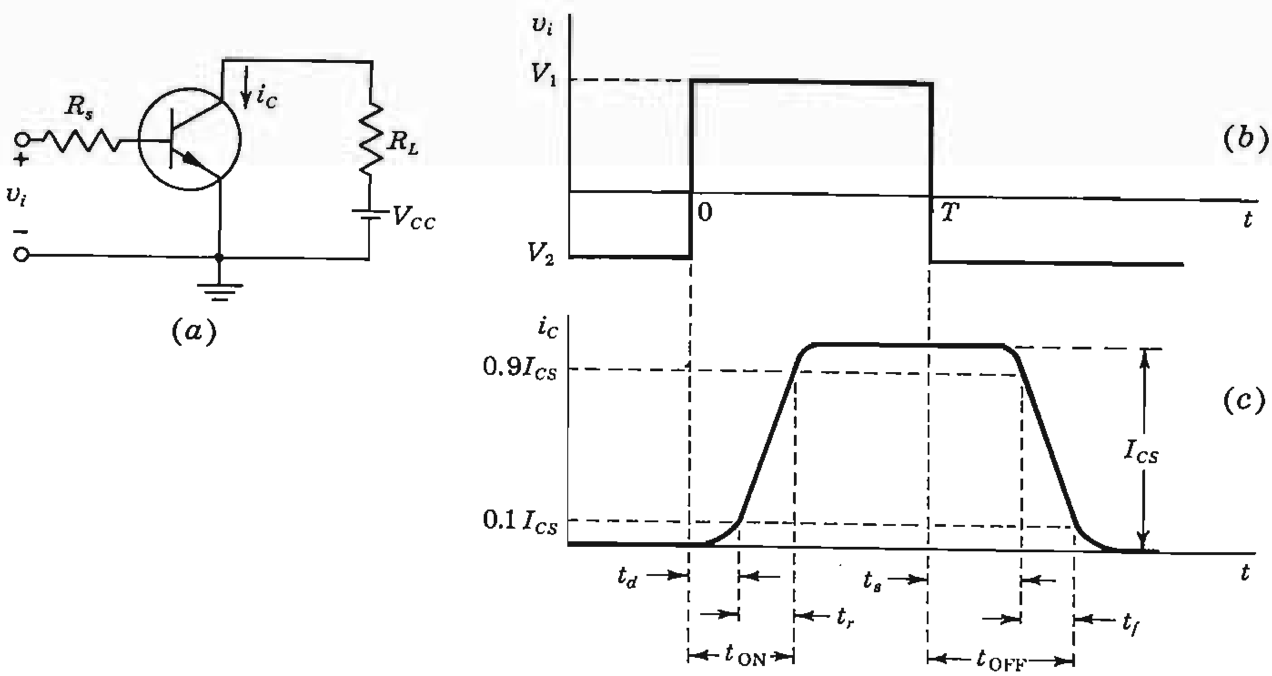


Fig. 6-9 The pulse waveform in (*b*) drives the transistor in (*a*) from cutoff to saturation and back again. (c) The collector-current response to the driving input pulse.

time t_{OFF} is defined as the sum of the storage and fall times, $t_{\text{OFF}} \equiv t_s + t_f$. We shall consider now the physical reasons for the existence of each of these times. The actual calculation of the time intervals (t_d, t_r, t_s, and t_f) is complex, and the reader is referred to Ref. 2. Numerical values of delay time, rise time, storage time, and fall time for the Texas Instruments *n-p-n* epitaxial planar silicon transistor 2N3830 under specified conditions can be as low as $t_d = 10$ ns, $t_r = 50$ ns, $t_s = 40$ ns, and $t_f = 30$ ns.

The Delay Time Three factors contribute to the delay time. First, when the driving signal is applied to the transistor input, a nonzero time is required to charge up the emitter-junction transition capacitance so that the transistor may be brought from cutoff to the active region. Second, even when the transistor has been brought to the point where minority carriers have begun to cross the emitter junction into the base, a time interval is required before these carriers can cross the base region to the collector junction and be recorded as collector current. Finally, some time is required for the collector current to rise to 10 percent of its maximum.

Rise Time and Fall Time The rise time and the fall time are due to the fact that, if a base-current step is used to saturate the transistor or return it from saturation to cutoff, the transistor collector current must traverse the active region. The collector current increases or decreases along an exponential curve whose time constant τ_r can be shown to be given by $\tau_r = h_{FE}(C_c R_c + 1/\omega_T)$, where C_c is the collector transition capacitance and ω_T is the radian frequency at which the current gain is unity (Sec. 11-3).

Storage Time The failure of the transistor to respond to the trailing edge of the driving pulse for the time interval t_s (indicated in Fig. 6-9c) results from the fact that a transistor in saturation has a saturation charge of excess minority carriers stored in the base. The transistor cannot respond until this

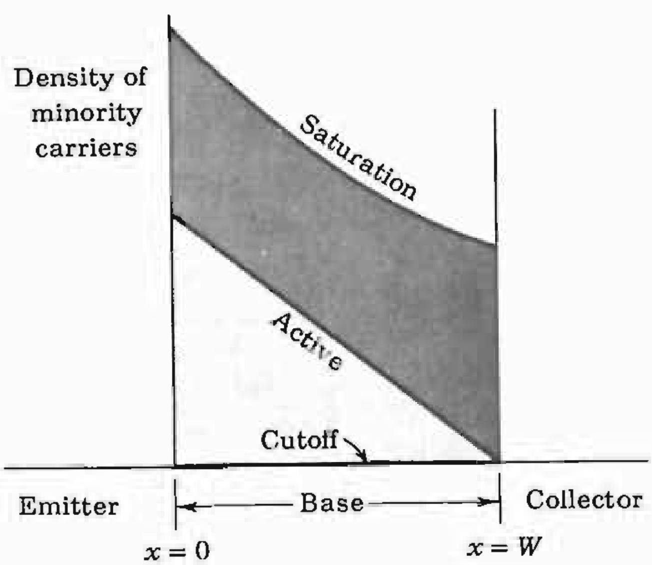


Fig. 6-10 Minority-carrier concentration in the base for cutoff, active, and saturation conditions of operation.

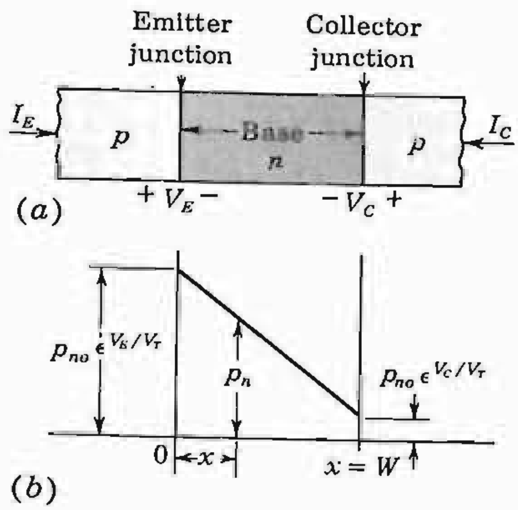


Fig. 6-11 The minority-carrier density in the base region. The concentrations at $x = 0$ and $x = W$ are governed by the law of the junction [(Eq. 3-4)].

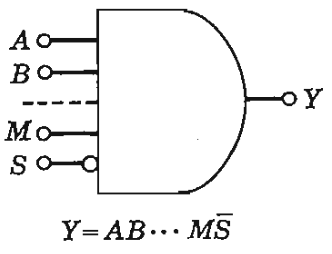
saturation excess charge has been removed. The stored charge density in the base is indicated in Fig. 6-10 under various operating conditions.

The concentration of minority carriers in the base region decreases linearly from $p_{no}\epsilon^{V_E/V_T}$ at $x = 0$ to $p_{no}\epsilon^{V_C/V_T}$ at $x = W$, as indicated in Fig. 6-11*b*. In the cutoff region, both V_E and V_C are negative, and p_n is almost zero everywhere. In the active region, V_E is positive and V_C negative, so that p_n is large at $x = 0$ and almost zero at $x = W$. Finally, in the saturation region, where V_E and V_C are both positive, p_n is large everywhere, and hence a large amount of minority-carrier charge is stored in the base. These densities are pictured in Fig. 6-10.

Consider that the transistor is in its saturation region and that at $t = T$ an input step is used to turn the transistor off, as in Fig. 6-9. Since the turnoff process cannot begin until the abnormal carrier density (the heavily shaded area of Fig. 6-10) has been removed, a relatively long storage delay time t_s may elapse before the transistor responds to the turnoff signal at the input. In an extreme case this storage-time delay may be two or three times the rise or fall time through the active region. It is clear that, when transistor switches are to be used in an application where speed is at a premium, it is advantageous to reduce the storage time. By adding the capacitor C across the base resistor (Fig. 6-7), an impulsive current will flow out of the base at the time T at the end of the pulse. If C is properly chosen,[3] this impulsive current will instantaneously reduce t_s to zero. A method for preventing a transistor from saturating, and thus eliminating storage time, is given in Sec. 7-13.

6-6 THE INHIBIT (ENABLE) OPERATION

A NOT circuit preceding one terminal (S) of an AND gate acts as an *inhibitor*. This modified AND circuit implements the logical statement. *If $A = 1$, $B = 1$, . . . , $M = 1$, then $Y = 1$ provided that $S = 0$. However, if $S = 1$, then the*



(a)

	Input			Output
	A	B	S	Y
1	0	0	0	0
2	0	1	0	0
3	1	0	0	0
4	1	1	0	1
5	0	0	1	0
6	0	1	1	0
7	1	0	1	0
8	1	1	1	0

(b)

Fig. 6-12 (a) The logic block and Boolean expression for an AND with an enable terminal S. (b) The truth table for $Y = AB\bar{S}$. The column on the left numbers the eight possible input combinations.

coincidence of A, B, . . . , M is inhibited, and $Y = 0$. Such a configuration is also called an *anticoincidence* circuit. The logical block symbol is drawn in Fig. 6-12*a*, together with its Boolean equation. The equation is to be read "Y equals A *and* B *and . . . and* M *and not* S." The truth table for a three-input AND gate with one inhibitor terminal (S) is given in Fig. 6-12*b*.

The terminal S is also called a *strobe* or an *enable input*. The enabling bit $S = 0$ allows the gate to perform its AND logic, whereas the inhibiting bit $S = 1$ causes the output to remain at $Y = 0$, independently of the values of the input bits.

A combination of the AND circuit of Fig. 6-5*b* and the INVERTER of Fig. 6-8 satisfying the logic given in the truth table (Fig. 6-12*b*) is indicated in Fig. 6-13. If either input A or B or both are in the 0 state, $V(0) = 0$ V, then at least one of the diodes $D1$ or $D2$ conducts and clamps the output to 0 V, or $Y = 0$. This argument verifies all items in the truth table except lines 4 and 8. Consider now the situation where a coincidence occurs at A and B. If C is in the 0 state, then Q is cut off, and the output of the NOT circuit is $\bar{C} = 1$ (12 V). Hence all

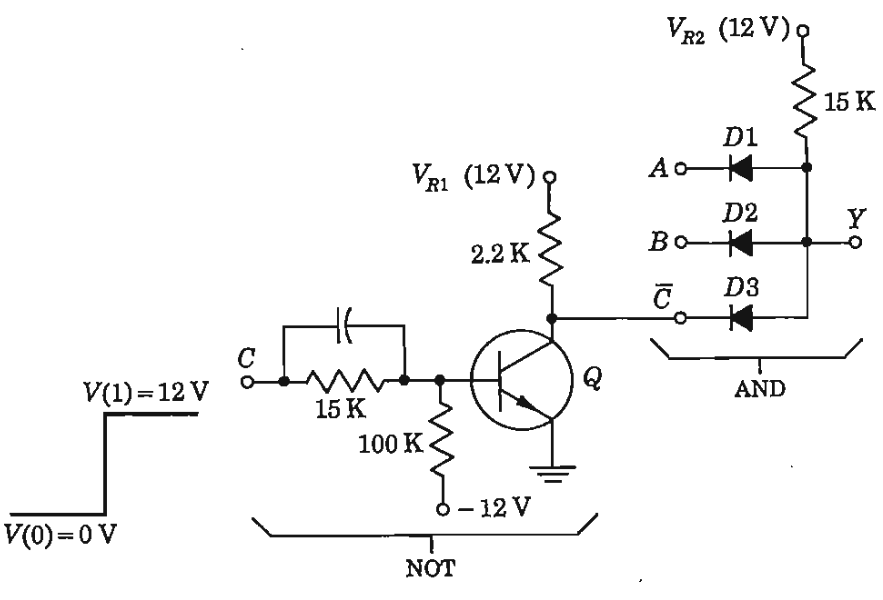


Fig. 6-13 A positive-logic AND circuit with a negation input terminal.

three diodes are reverse-biased and the output rises to 12 V, or $Y = 1$, which verifies line 4 of the truth table. (If A, B, V_{R1}, and V_{R2} are not all equal to the same voltage, the output will rise to the smallest of these values.) Finally, consider the condition in line 8 of Fig. 6-12*b*. If C is in the 1 state, then Q is driven into saturation, and the output of the transistor drops to 0 V (ideally). Hence $\bar{C} = 0$, $D3$ conducts, and $Y = 0$, which indeed is the logic in the last row of the truth table.

It is possible to have a two-input AND, one terminal of which is inhibiting. This circuit satisfies the logic: "The output is true (1) if input A is true (1) provided that B is not true (0) [or equivalently, provided that B is false (0)]." Another possible configuration is an AND with more than one inhibit terminal.

In a dynamic system, if an inhibit pulse is to allow more of the signal to be transmitted through the gate, it is necessary that the inhibit pulse begin earlier and last longer than the signal pulses.

6-7 THE EXCLUSIVE OR CIRCUIT

An EXCLUSIVE OR gate obeys the definition: *The output of a two-input* EXCLUSIVE OR *assumes the* 1 *state if one and only one input assumes the* 1 *state.* The standard symbol for an EXCLUSIVE OR is given in Fig. 6-14*a* and the truth table in Fig. 6-14*b*. The circuit of Sec. 6-2 is referred to as an INCLUSIVE OR if it is desired to distinguish it from the EXCLUSIVE OR.

The above definition is equivalent to the statement: "If $A = 1$ or $B = 1$ but not simultaneously, then $Y = 1$." In Boolean notation,

$$Y = (A + B)(\overline{AB}) \tag{6-20}$$

This function is implemented in logic diagram form in Fig. 6-15*a*.

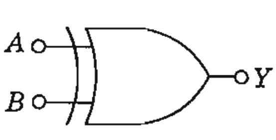
A second logic statement equivalent to the definition of the EXCLUSIVE OR is the following: "If $A = 1$ and $B = 0$, or if $B = 1$ and $A = 0$, then $Y = 1$." The Boolean expression is

$$Y = A\bar{B} + B\bar{A} \tag{6-21}$$

The block diagram which satisfies this logic is indicated in Fig. 6-15*b*.

An EXCLUSIVE OR is employed within the arithmetic section of a computer. Another application is as an *inequality comparator, matching circuit,* or *detector* because, as can be seen from the truth table, $Y = 1$ only if $A \neq B$. This

Fig. 6-14 (*a*) The standard symbol for an EXCLUSIVE OR gate and its Boolean expression; (*b*) the truth table.



$Y = A \oplus B$

(*a*)

Input		Output
A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

(*b*)

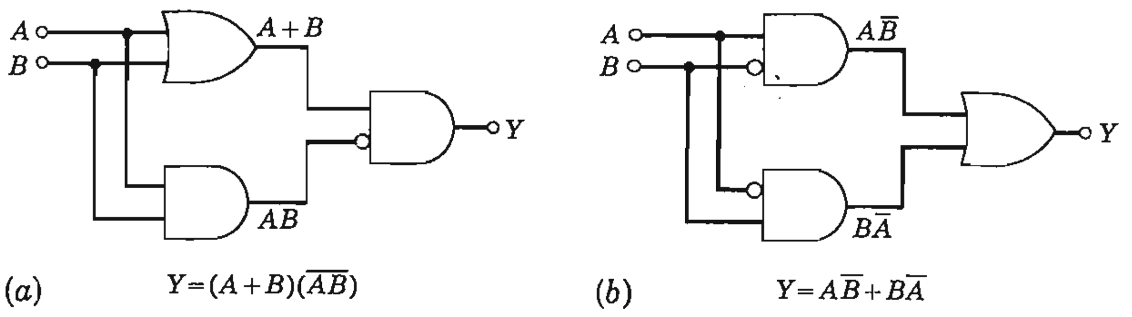


Fig. 6-15 Two logic block diagrams for the EXCLUSIVE OR gate.

property is used to check for the inequality of two bits. If bit A is not identical with bit B, then an output is obtained. Equivalently, "If A and B are both 1 or if A and B are both 0, then no output is obtained, and $Y = 0$." This latter statement may be put into Boolean form as

$$Y = \overline{AB + \bar{A}\bar{B}} \tag{6-22}$$

This equation leads to a third implementation for the EXCLUSIVE OR block, which is indicated by the logic diagram of Fig. 6-16*a*. An *equality detector* gives an output $Z = 1$ if A and B are both 1 or if A and B are both 0, and hence

$$Z = \bar{Y} = AB + \bar{A}\bar{B} \tag{6-23}$$

where use was made of Eq. (6-16). If the output Z is desired, the negation in Fig. 6-16*a* may be omitted or an additional inverter may be cascaded with the output of the EXCLUSIVE OR.

A fourth possibility for this gate is

$$Y = (A + B)(\bar{A} + \bar{B}) \tag{6-24}$$

which may be verified from the definition or from the truth table. This logic is depicted in Fig. 6-16*b*.

We have demonstrated that there often are several ways to implement a logical circuit. In practice one of these may be realized more advantageously than the others. Boolean algebra is sometimes employed for manipulating a logic equation so as to transform it into a form which is better from the point

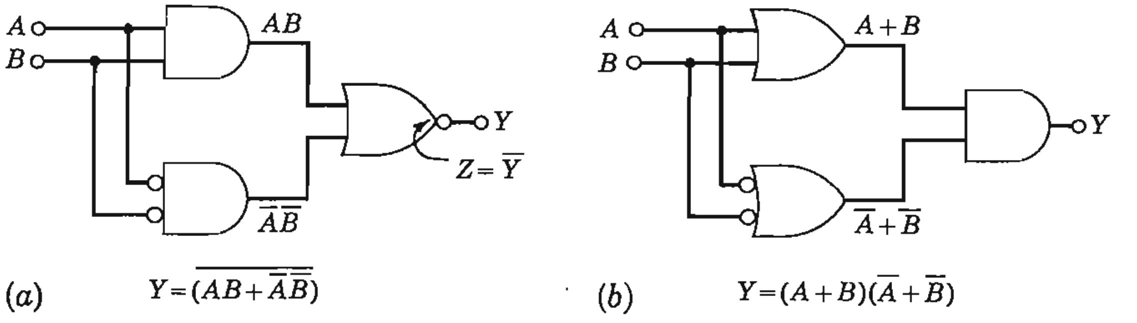


Fig. 6-16 Two additional logic block diagrams for the EXCLUSIVE OR gate.

of view of implementation in hardware. In the next section we shall verify through the use of Boolean algebra that the four expressions given above for the EXCLUSIVE OR are equivalent.

Two-level Logic Digital design often calls for several gates (AND, OR, or combinations of those) feeding into an OR (or AND) gate. Such a combination is known as *two-level (or two-wide) logic*. The EXCLUSIVE OR circuits of Figs. 6-15 and 6-16 are examples of two-level logic. In the discussion of digital systems in Chap. 17 it is found that the most useful logic array consists of several ANDs which feed an OR which is followed by a NOT gate. This cascade of gates is called an AND-OR-INVERT (AOI) configuration. The detailed circuit topology for an AOI is given in Fig. 17-2.

6-8 DE MORGAN'S LAWS

The statement "If and only if all inputs are true (1), then the output is true (1)" is logically equivalent to the statement "If at least one input is false (0), then the output is false (0)." In Boolean notation this equivalence is written

$$ABC \cdots = \overline{\bar{A} + \bar{B} + \bar{C} + \cdots} \tag{6-25}$$

If we take the complement of both sides of this equation and use Eq. (6-16), we obtain

$$\overline{ABC \cdots} = \bar{A} + \bar{B} + \bar{C} + \cdots \tag{6-26}$$

This equation and its dual,

$$\overline{A + B + C + \cdots} = \bar{A}\bar{B}\bar{C} \cdots \tag{6-27}$$

(which may be proved in a similar manner), are known as De Morgan's laws. These complete the list of basic Boolean identities. For easy future reference, all these relationships are summarized in Table 6-4.

With the aid of Boolean algebra we shall now demonstrate the equivalence of the four EXCLUSIVE OR circuits of the preceding section. Using Eq. (6-26), it is immediately clear that Eq. (6-21) is equivalent to Eq. (6-24). Now the latter equation can be expanded with the aid of Table 6-4 as follows:

$$(A + B)(\bar{A} + \bar{B}) = A\bar{A} + B\bar{A} + A\bar{B} + B\bar{B} = B\bar{A} + A\bar{B} \tag{6-28}$$

This result shows that the EXCLUSIVE OR of Eq. (6-21) is equivalent to that of Eq. (6-24). Finally, applying Eq. (6-27) to Eq. (6-22) gives

$$\overline{AB + \bar{A}\bar{B}} = (\overline{AB})(\overline{\bar{A}\bar{B}}) \tag{6-29}$$

From Eq. (6-26), we have

$$(\overline{AB})(\overline{\bar{A}\bar{B}}) = (\bar{A} + \bar{B})(\bar{\bar{A}} + \bar{\bar{B}}) = (\bar{A} + \bar{B})(A + B) \tag{6-30}$$

TABLE 6-4 Summary of basic Boolean identities

Fundamental laws

OR	AND	NOT
$A + 0 = A$	$A0 = 0$	$A + \bar{A} = 1$
$A + 1 = 1$	$A1 = A$	$A\bar{A} = 0$
$A + A = A$	$AA = A$	$\bar{\bar{A}} = A$
$A + \bar{A} = 1$	$A\bar{A} = 0$	

Associative laws

$$(A + B) + C = A + (B + C) \qquad (AB)C = A(BC)$$

Commutative laws

$$A + B = B + A \qquad AB = BA$$

Distributive law

$$A(B + C) = AB + AC$$

De Morgan's laws

$$\overline{AB \cdots} = \bar{A} + \bar{B} + \cdots$$

$$\overline{A + B + \cdots} = \bar{A}\bar{B} \cdots$$

Auxiliary identities

$$A + AB = A \qquad A + \bar{A}B = A + B$$

$$(A + B)(A + C) = A + BC$$

where use is made of the identity $\bar{\bar{A}} = A$. Comparing Eqs. (6-29) and (6-30) shows that the EXCLUSIVE OR of Eq. (6-22) is equivalent to that of Eq. (6-24).

With the aid of De Morgan's law we can show that *an* AND *circuit for positive logic also functions as an* OR *gate for negative logic.* Let Y be the output and $A, B, \ldots, N$ be the inputs to a positive AND so that

$$Y = AB \cdots N \tag{6-31}$$

Then, by Eq. (6-26),

$$\bar{Y} = \bar{A} + \bar{B} + \cdots + \bar{N} \tag{6-32}$$

If the output and all inputs of a circuit are complemented so that a 1 becomes a 0 and vice versa, then positive logic is changed to negative logic (refer to Fig. 6-1). Since Y and $\bar{Y}$ represent the *same* output terminal, A and $\bar{A}$ the *same* input terminal, etc., the circuit which performs the positive AND logic in Eq. (6-31) also operates as the negative OR gate of Eq. (6-32). Similar reasoning is used to verify that the same circuit is either a negative AND or a positive OR, depending upon how the binary levels are defined. We verified this result for diode logic in Sec. 6-3, but the present proof is independent of how the circuit is implemented.

It should now be clear that it is really not necessary to use all three connectives OR, AND, and NOT. The OR and the NOT are sufficient because, from the De Morgan law of Eq. (6-25), the AND can be obtained from the OR and the NOT, as is indicated in Fig. 6-17*a*. Similarly, the AND and the NOT may be chosen as the basic logic circuits, and from the De Morgan law of Eq. (6-27), the OR may be constructed as shown in Fig. 6-17*b*. This figure makes clear

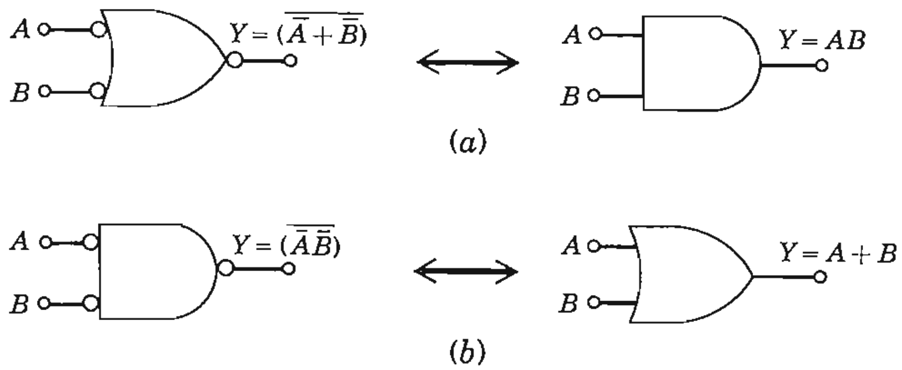


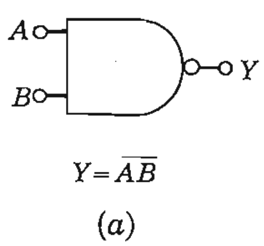
Fig. 6-17 (a) An OR is converted into an AND by inverting all inputs and also the output. (b) An AND becomes an OR if all inputs and the output are complemented.

once again that an OR (AND) circuit negated at input and output performs the AND (OR) logic.

6-9 THE NAND AND NOR DIODE-TRANSISTOR LOGIC (DTL) GATES

In Fig. 6-15*a* the negation before the second AND could equally well be put at the output of the first AND without changing the logic. Such an AND-NOT sequence is also present in Fig. 6-17*b* and in many other logic operations. This negated AND is called a NOT-AND, or a NAND, gate. The logic symbol, Boolean expression, and truth table for the NAND are given in Fig. 6-18. The NAND may be implemented by placing a transistor NOT circuit *after* the diode AND in Fig. 6-13. Such a transposition is shown in Fig. 6-19. Circuits involving diodes and transistors as in Fig. 6-19 are called *diode-transistor logic* (DTL) *gates.*

EXAMPLE (*a*) Verify that the circuit of Fig. 6-19 is a positive NAND for the binary levels 0 and 12 V. Neglect source impedance and junction saturation voltages and diode voltages in the forward direction. Find the minimum h_{FE}. (*b*) If the drop across a conducting diode is 0.7 V and if the sum of source and diode resistances is 1 K, is the NAND logic satisfied? (*c*) Will the circuit operate



Input A	Input B	Output Y
0	0	1
0	1	1
1	0	1
1	1	0

(*b*)

Fig. 6-18 (a) The logic symbol and Boolean expression for a two-input NAND gate; (b) the truth table.

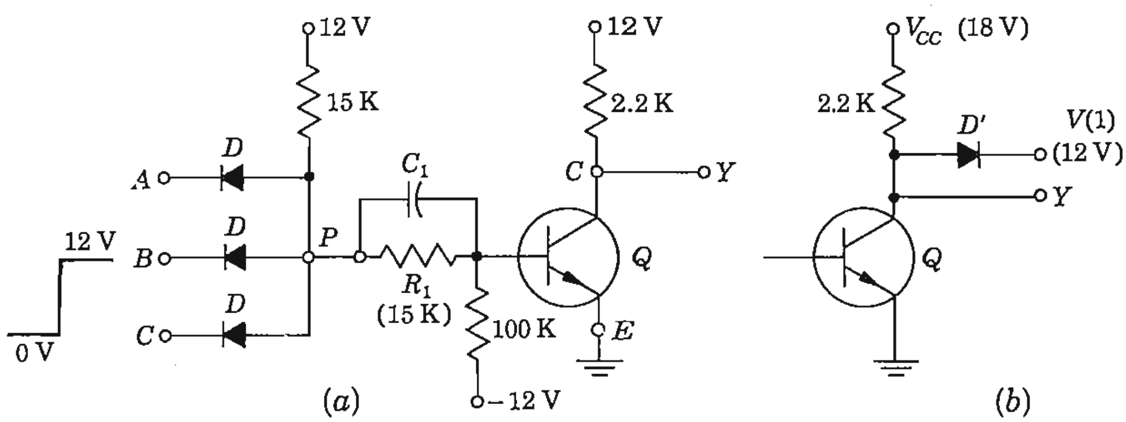


Fig. 6-19 (*a*) A three-input positive NAND (or negative NOR) gate; (*b*) a collector catching or clamping diode may be used to improve the characteristics of the output circuit.

properly if the inputs are obtained from the outputs of similar NAND gates? Assume silicon transistors and diodes and neglect collector saturation resistance and diode forward resistance.

Solution *a*. If any input is at 0 V, then the junction point P of the diodes is at 0 V because a diode conducts and clamps this point to $V(0) = 0$. The base voltage of the transistor is then

$$V_B = -(12)\left(\frac{15}{112}\right) = -1.56 \text{ V}$$

Hence Q is cut off and Y is at 12 V, or $Y = 1$. This result confirms the first three rows of the truth table of Fig. 6-18*b*.

If all inputs are at $V(1) = 12$ V, assume that all diodes are reverse-biased and that the transistor is in saturation. We shall now verify that these assumptions are indeed correct. If Q is in saturation, then with $V_{BE} = 0$, the voltage at P is $(12)(\frac{15}{30}) = 6$ V. Hence, with 12 V at each input, all diodes are reverse-biased by 6 V. Since the diodes are nonconducting, the two 15-K resistors are in series and the base current of Q is

$$\frac{12}{30} - \frac{12}{100} = 0.40 - 0.12 = 0.28 \text{ mA}$$

Since the collector current is

$$I_C = \frac{12}{2.2} = 5.45 \text{ mA} \qquad \text{and} \qquad (h_{FE})_{\min} = \frac{5.45}{0.28} = 19$$

then Q will indeed be in saturation if $h_{FE} \geq 19$. Under these circumstances the output is at ground, or $Y = 0$. This result confirms the last row of the truth table.

b. The transistor must be OFF if at least one input is at 0. The worst case occurs when all diodes except one are reverse-biased, because then the voltage at P is a maximum. For this case the Thévenin's equivalent from P to ground is,

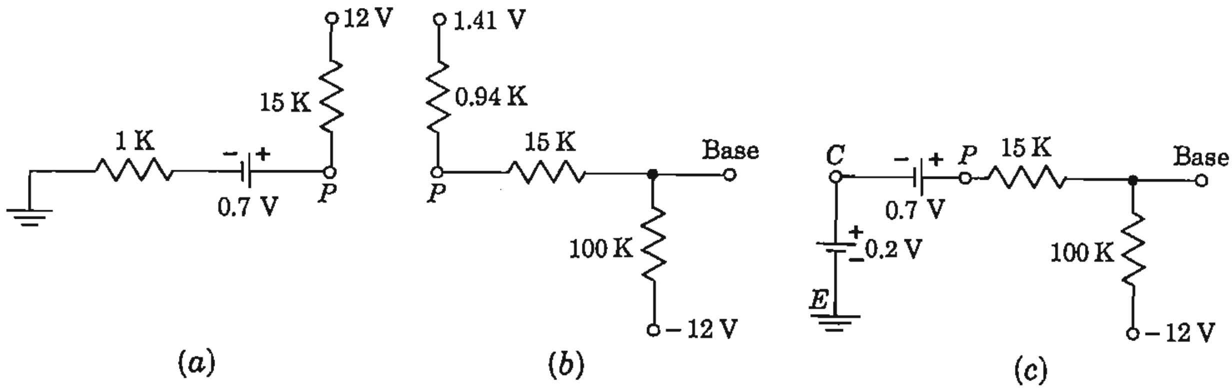


Fig. 6-20 Relating to the calculation of the base voltage of the transistor in the circuit of Fig. 6-19.

from Fig. 6-20a, a voltage of

$$(12)\left(\frac{1}{16}\right) + (0.7)\left(\frac{15}{16}\right) = 0.75 + 0.66 = 1.41 \text{ V}$$

in series with a resistance of $(1)(\frac{15}{16}) = 0.94$ K. The open-circuit voltage at the base of the transistor is, from Fig. 6-20b,

$$V_B = -(12)\left(\frac{15.9}{115.9}\right) + (1.41)\left(\frac{100}{115.9}\right) = -1.65 + 1.21 = -0.44 \text{ V}$$

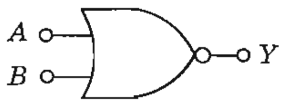
This voltage is more than adequate to reverse-bias Q, and hence NAND logic is satisfied.

c. If the inputs are high, the situation is exactly as in part *a*. With respect to keeping the base node at a low voltage when there is no coincidence, the worst situation occurs when all but one input are high. The low input now comes from a transistor in saturation, and $V_{CE,\text{sat}} \approx 0.2$ V. The open-circuit voltage at the base of Q is, from Fig. 6-20c,

$$V_B = -(12)\left(\frac{15}{115}\right) + (0.9)\left(\frac{100}{115}\right) = -0.78 \text{ V}$$

which cuts off Q and $Y = 1$, as it should.

If we neglect the inherent speed limitations of the transistor the rise time of the output, when Q is cut off, depends upon the shunt capacitance C_s and the collector resistance R_c. If V_{CC} is increased, then for fixed values of C_s and R_c, less time is required to reach the particular voltage at which the next stage switches (is driven into saturation). If such an increased value of V_{CC} [$> V(1)$] is used, a collector clamping diode is often added, as indicated in Fig. 6-19b. This diode limits the collector voltage of Q to $V(1)$ and also prevents the following stage from being driven too heavily into saturation. Secondarily, the diode helps to reduce the time to discharge C_s when Q is driven into saturation by providing a lower collector starting voltage.



$Y=\overline{A+B}$

(a)

Input A	Input B	Output Y
0	0	1
0	1	0
1	0	0
1	1	0

(b)

Fig. 6-21 (a) The logic symbol and Boolean expression for a two-input NOR gate; (b) the truth table.

A NOR Gate A negation following an OR is called a NOT-OR, or a NOR gate. The logic symbol, Boolean expression, and truth table for the NOR are given in Fig. 6-21. A positive NOR circuit is implemented in Fig. 6-22*a* by a cascade of a diode OR and a transistor INVERTER.

In Fig. 6-22*a* the base supply $-V_{BB}$ may also be used as the reference voltage V_R for the OR, and hence V_R and R may be omitted from the circuit. Such a simplified configuration is indicated in Fig. 6-22*b* for the specific binary levels $V(0) = 0$ and $V(1) = 12$ V. We can readily confirm that this circuit obeys NOR logic. If all inputs are in the 0 state, all diodes conduct and the input to the inverter is 0 V. If any input is high, the diode connected to this input conducts and all other diodes are reverse-biased. The voltage at the diode node P is now $V(1) = 12$ V. Hence, from input to point P, the OR function has been satisfied. Since from P to the output we have an inverter identical with that in Fig. 6-8, no further calculations are necessary to justify NOR operation. The direct connection from the junction of the OR diodes to the external pin C is convenient for expanding the number of inputs by adding more diodes as needed.

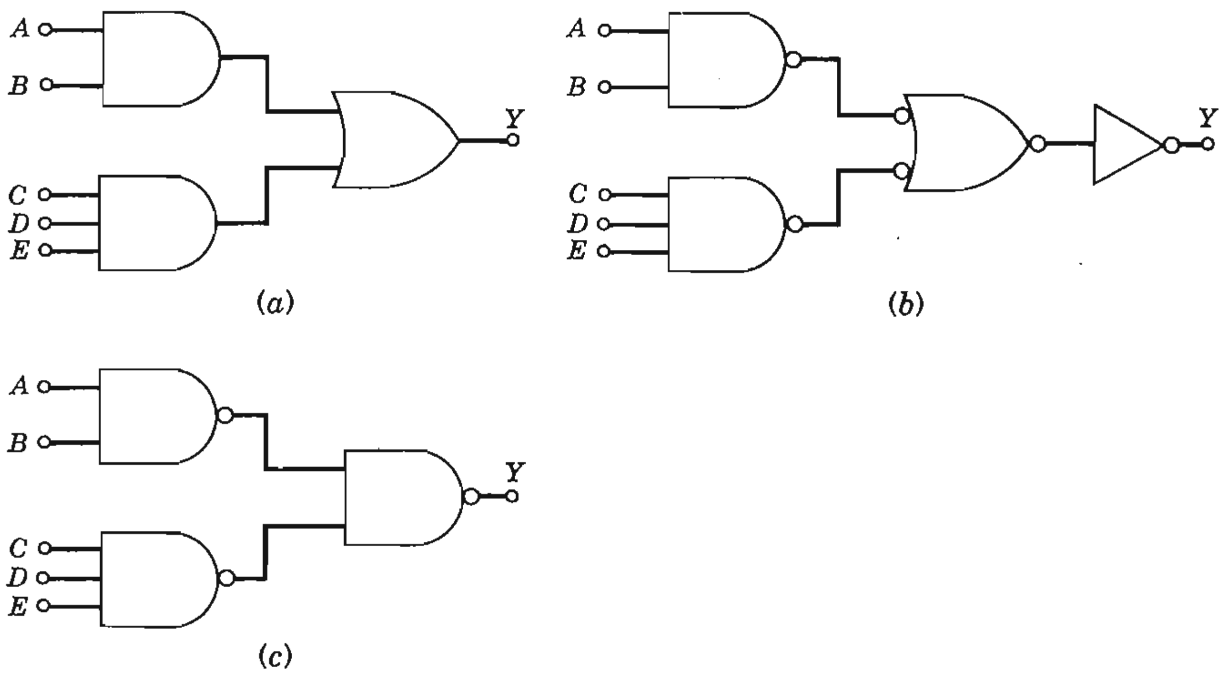


Fig. 6-22 A two-level AND-OR is equivalent to a NAND-NAND configuration.

The circuits of Figs. 6-19 and 6-22 employ *diode-transistor logic* (DTL). The NAND and NOR may also be implemented in other configurations, as is indicated in Secs. 6-12 and 6-14. With the aid of De Morgan's laws, it can be shown that, regardless of the hardware involved, a positive NAND is also a negative NOR, whereas a negative NAND may equally well be considered a positive NOR.

It is clear that a single input NAND is a NOT. Also, a NAND followed by a NOT is an AND. In Sec. 6-8 it is pointed out that all logic can be performed by using only the two connectives AND and NOT. Therefore we now conclude that, by repeated use of the NAND circuit alone, any logical function can be carried out. A similar argument leads equally well to the result that all logic can be performed by using only the NOR circuit.

EXAMPLE Verify that two-level AND-OR topology is equivalent to a NAND-NAND system.

Solution The AND-OR logic is indicated in Fig. 6-22*a*. Since $X = \bar{\bar{X}}$, then inverting the output of an AND and simultaneously negating the input to the following OR does not change the logic. These modifications are made in Fig. 6-22*b*. We have also negated the output of the OR gate and, at the same time, have added an INVERTER to Fig. 6-22*b*, so that once again the logic is unaffected. An OR gate negated at each terminal is an AND circuit (Fig. 6-17*a*). Since an AND followed by an INVERTER is a NAND then Fig. 6-22*c* is equivalent to Fig. 6-22*b*. Hence, the NAND-NAND of Fig. 6-22*c* is equivalent to the AND-OR of Fig. 6-22*a*.

If any of the inputs in Fig. 6-22 are obtained from the output of another gate then the resultant topology is referred to as *three-level logic*.

6-10 MODIFIED (INTEGRATED-CIRCUIT) DTL GATES[5]

Most logic gates are fabricated as an *integrated circuit* (IC). This process is described in the next chapter, where it is demonstrated that all transistors, diodes, resistors, and capacitors in a fairly complicated circuit may be shaped within a tiny chip of single-crystal silicon (approximately 50 mil by 50 mil in surface area and 1 mil thick). It turns out that large values of resistance (above 30 K) and of capacitance (above 100 pF) cannot be fabricated economically. On the other hand, transistors and diodes may be constructed very inexpensively. In view of these facts, the NAND gate of Fig. 6-19 is modified for integrated-circuit implementation by eliminating the capacitor C_1, reducing the resistance values drastically, and using diodes or transistors to replace resistors wherever possible. At the same time the power-supply requirements are simplified so that only a single 5-V supply is used (instead of the three separate voltages of Fig. 6-19*b*). The resulting circuit is indicated in Fig. 6-23.

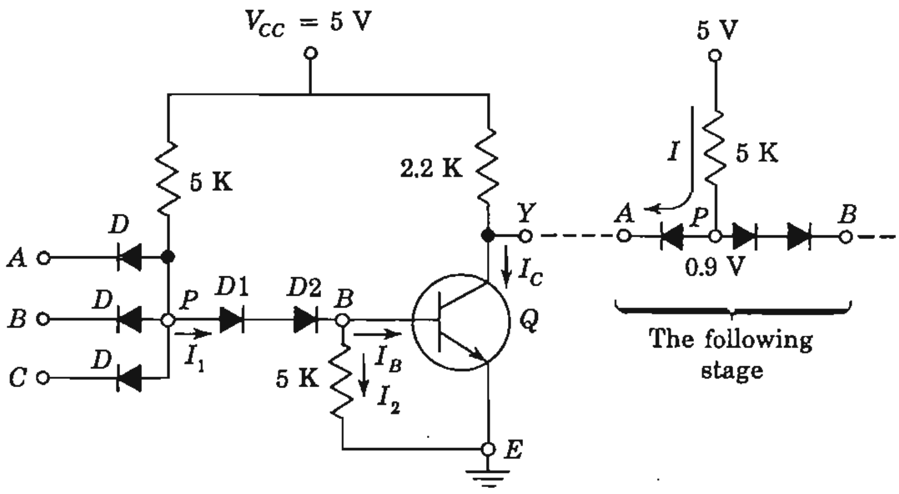


Fig. 6-23' An integrated positive DTL NAND gate.

The operation of this positive NAND gate is easily understood qualitatively. If at least one of the inputs is low (the 0 state), the diode D connected to this input conducts and the voltage V_P at point P is low. Hence diodes $D1$ and $D2$ are nonconducting, $I_B = 0$, and the transistor is OFF. Therefore the output of Q is high and Y is in the 1 state. This logic satisfies the first three rows of the truth table in Fig. 6-18. Consider now the case where all inputs are high (1) so that all input diodes D are cut off. Then V_P tries to rise toward V_{CC}, and a base current I_B results. If I_B is sufficiently large, Q is driven into saturation and the output Y falls to its low (0) state, thus satisfying the fourth row of the truth table.

This NAND gate is considered quantitatively in the following illustrative example. The necessity for using two diodes $D1$ and $D2$ in series is explained. False logic can be caused by switching transients, power-supply noise spikes, coupling between leads, etc. The noise margins that this circuit can tolerate are calculated below

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EXAMPLE (*a*) For the transistor in Fig. 6-23 assume (Table 5-1) that $V_{BE,\text{sat}} = 0.8$ V, $V_\gamma = 0.5$ V, and $V_{CE,\text{sat}} = 0.2$ V. The drop across a conducting diode is 0.7 V and V_γ (diode) = 0.6 V. The inputs of this switch are obtained from the outputs of similar gates. Verify that the circuit functions as a positive NAND and calculate $(h_{FE})_{\text{min}}$. (*b*) Will the circuit operate properly if $D2$ is not used? (*c*) If all inputs are high, what is the magnitude of the noise voltage at the input which will cause the gate to malfunction? (*d*) Repeat part *c* if at least one input is low. Assume, for the moment, that Q is not loaded by a following stage.

Solution *a*. The logic levels are $V_{CE,\text{sat}} = 0.2$ V for the 0 state and $V_{CC} = 5$ V for the 1 state. If at least one input is in the 0 state, its diode conducts and $V_P = 0.2 + 0.7 = 0.9$ V. Since, in order for $D1$ and $D2$ to be conducting, a voltage of $(2)(0.7) = 1.4$ V is required, these diodes are cut off, and $V_{BE} = 0$. Since the cutin voltage of Q is $V_\gamma = 0.5$ V, then Q is OFF, the output rises to 5 V, and $Y = 1$. This confirms the first three rows of the NAND truth table.

If all inputs are at $V(1) = 5$ V, then we shall assume that all input diodes are OFF, that $D1$ and $D2$ conduct, and that Q is in saturation. If these conditions are true, the voltage at P is the sum of two diode drops plus $V_{BE,\text{sat}}$, or $V_P = 0.7 + 0.7 + 0.8 = 2.2$ V. The voltage across each input diode is $5 - 2.2 = 2.8$ V in the reverse direction, thus justifying the assumption that D is OFF. We now find $(h_{FE})_{\min}$ to put Q into saturation.

$$I_1 = \frac{V_{CC} - V_P}{5} = \frac{5 - 2.2}{5} = 0.56 \text{ mA}$$

$$I_2 = \frac{V_{BE,\text{sat}}}{5} = \frac{0.8}{5} = 0.16 \text{ mA}$$

$$I_B = I_1 - I_2 = 0.56 - 0.16 = 0.40 \text{ mA}$$

$$I_C = \frac{V_{CC} - V_{CE,\text{sat}}}{2.2} = \frac{5 - 0.2}{2.2} = 2.18 \text{ mA}$$

and

$$(h_{FE})_{\min} = \frac{I_C}{I_B} = \frac{2.18}{0.40} = 5.5$$

If $h_{FE} > (h_{FE})_{\min}$, then $Y = V(0)$ for all inputs at $V(1)$, thus verifying the last line in the truth table in Fig. 6-18.

b. If at least one input is at $V(0)$, then $V_P = 0.2 + 0.7 = 0.9$ V. Hence, if only one diode $D1$ is used between P and B, then $V_{BE} = 0.9 - 0.6 = 0.3$ V, where 0.6 V represents the diode cutin voltage. Since the cutin base voltage is $V_\gamma = 0.5$ V, then theoretically Q is cut off. However, this is not a very conservative design because a small (>0.2 V) spike of noise will turn Q ON. An even more conservative design uses three diodes in series, instead of the two indicated in Fig. 6-23.

c. If all inputs are high, then from part *a*, $V_P = 2.2$ V and each input diode is reverse-biased by 2.8 V. A diode starts to conduct when it is forward-biased by 0.6 V. Hence a negative noise spike in excess of $2.8 + 0.6 = 3.4$ V must be present at the input before the circuit manfunctions. Such a large noise voltage is improbable.

d. If at least one input is low, then from part *a*, $V_P = 0.9$ V and Q is OFF. If a noise spike takes Q just into its active region, $V_{BE} = V_\gamma = 0.5$ and V_P must increase to $0.5 + 0.6 + 0.6 = 1.7$ V. Hence the noise margin is $1.7 - 0.9 = 0.8$ V. If only one diode $D1$ were used, the noise voltage would be reduced by 0.6 V (the drop across $D2$ at cutin) to $0.8 - 0.6 = 0.2$ V. This confirms the value obtained in part *b*.

Fan-out In the foregoing discussion we have unrealistically assumed that the NAND gate is unloaded. If it drives N similar gates, we say that the *fan-out* is N. The output transistor now acts as a *sink* for the current in the input to the gates it drives. In other words, when Q is in saturation ($Y = 0$), the input current I in Fig. 6-23 of a following stage adds to the collector

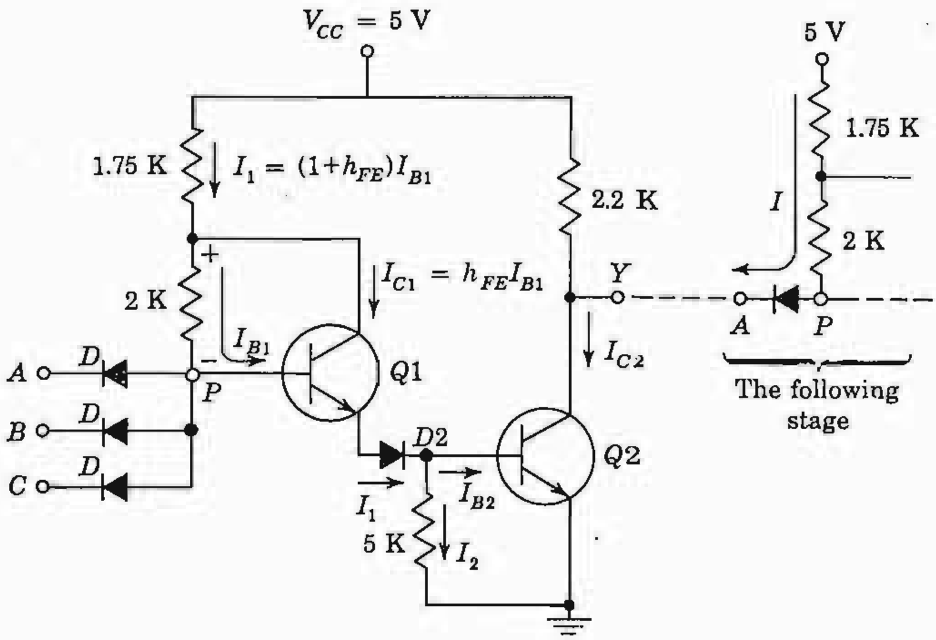


Fig. 6-24 A modified integrated positive DTL NAND gate with increased fan-out.

current of Q. Assume that all the input diodes to a following stage (which is now considered to be a *current source*) are high except the one driven by Q. Then the current in this diode is $I = (5 - 0.9)/5 = 0.82$ mA. This current is called a *standard load.* The total collector current of Q is now $I_C = 0.82N + 2.18$ mA, where 2.18 mA is the unloaded collector current found in part a of the preceding example. Since the base current is almost independent of loading, I_B remains at its previous value of 0.4 mA. If we assume a reasonable value for $(h_{FE})_{\min}$ of 30, the fan-out is given by $I_C = h_{FE}I_B$, or

$$I_C = 0.82N + 2.18 = (30)(0.40) = 12.0 \text{ mA} \tag{6-33}$$

and $N = 12$. Of course, the current rating of Q must exceed 12.0 mA if we are to drive 12 gates.

The fan-out may be increased considerably by replacing $D1$ by a transistor $Q1$, as indicated in Fig. 6-24. When $Q1$ is conducting, it is in its active region and *not* in saturation. This statement follows from the fact that the current in the 2-K resistance is in the direction to reverse-bias the collector junction of the *n-p-n* transistor $Q1$. Since the emitter current of $Q1$ supplies the base current of $Q2$, then $Q2$ is driven by a much higher base current than is Q in Fig. 6-23. For the same $(h_{FE})_{\min}$ of the output transistors in Figs. 6-23 and 6-24, it is clear that the latter circuit has the larger collector current, and hence the larger fan-out.

EXAMPLE If $(h_{FE})_{\min} = 30$, calculate the fan-out N for the NAND gate of Fig. 6-24. From Table 5-1, $V_{BE,\text{active}} = 0.7$ V.

Solution As with the circuit of Fig. 6-23, if any input is low, then $V_P = 0.9$ V, and both $Q1$ and $D2$ are OFF. Hence $V_{BE2} = 0$, $Q2$ is OFF, and $Y = 1$. If, however, all inputs are high, the input diodes are OFF, $Q2$ goes into saturation, and

$$V_P = V_{BE1,\text{active}} + V_{D2} + V_{BE2,\text{sat}} = 0.7 + 0.7 + 0.8 = 2.2 \text{ V}$$

Since $Q1$ is in its active region, $I_{C1} = h_{FE}I_{B1}$. As indicated in Fig. 6-24, the current in the 2-K resistor is I_{B1} (remember that each D is cut off), and the current in the 1.75-K resistor is $I_1 = I_{B1} + I_{C1} = (1 + h_{FE})I_{B1}$. Applying KVL between V_{CC} and V_P, we have, for $h_{FE} = 30$,

$$5 - 2.2 = (1.75)(31)I_{B1} + 2I_{B1} \tag{6-34}$$

or

$$I_{B1} = 0.050 \text{ mA} \qquad I_1 = (31)(0.050) = 1.55 \text{ mA}$$

$$I_2 = 0.8/5 = 0.16 \text{ mA} \qquad I_{B2} = 1.55 - 0.16 = 1.39 \text{ mA}$$

The unloaded collector current of $Q2$ is $(5 - 0.2)/2.2 = 2.18$ mA. For *each* gate which it drives, $Q2$ must sink a standard load of

$$I = \frac{5 - 0.7 - 0.2}{1.75 + 2} = 1.10 \text{ mA}$$

Since the maximum collector current is $h_{FE}I_{B2}$, then

$$I_{C2} = (30)(1.39) = 1.10N + 2.18 = 41.7 \text{ mA} \tag{6-35}$$

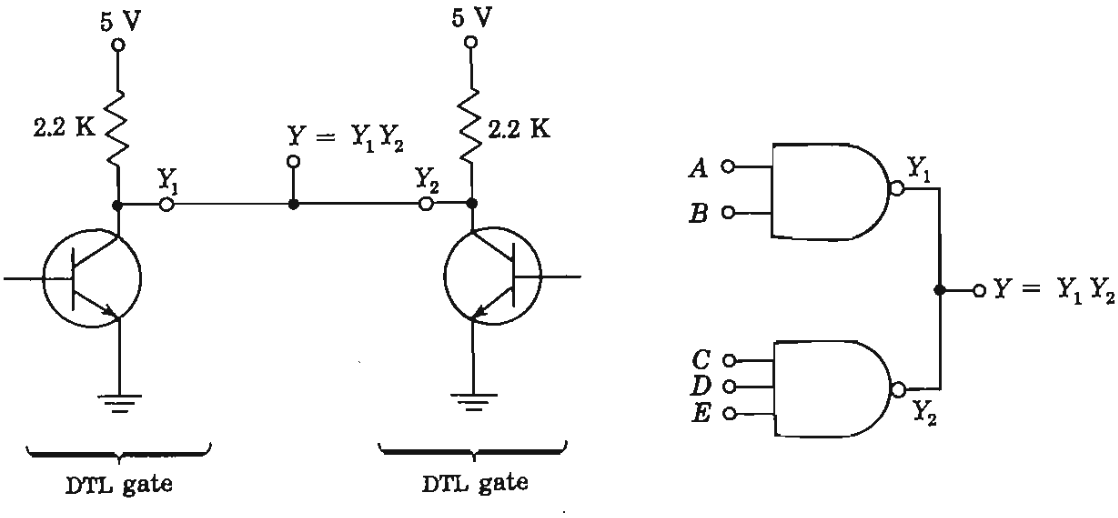
and $N = 36$. The fan-out has been increased to 36 for the same h_{FE} which resulted in only 12 for the circuit of Fig. 6-22.

The above calculation assumes that the current rating of $Q2$ is at least 41.7 mA. On the other hand, if $(I_{C2})_{\max}$ is limited to, say, 15 mA, then $1.10N + 2.18 = 15$, or $N \approx 11$. To drive these 11 gates requires that $h_{FE,\min} = I_{C2}/I_{B2} = 15/1.39 = 10.8$, which is a very small number.

The *fan-in M* of a logic gate gives the number of inputs to the switch. For example, in Fig. 6-24, $M = 3$.

Wired Logic It is possible to connect the outputs of several DTL gates together, as in Fig. 6-25, to perform additional logic without additional hardware. If positive NAND logic is under consideration, this connection is called a *wired*-AND, *phantom*-AND, *dotted*-AND, or *implied*-AND. Thus, if both $Y_1 = 1$ and $Y_2 = 1$, then $Y = 1$, whereas if $Y_1 = 0$ and/or $Y_2 = 0$, then $Y = 0$.

The circuit of Fig. 6-24 also represents a negative NOR gate, and connecting two outputs together as in Fig. 6-25 now represents negative wired-OR logic. If Y_1 and/or Y_2 is in the low state (which is now the 1 state), then Y is also in the low state ($Y = 1$), whereas if Y_1 and Y_2 are both high (the 0 state), then Y is also high ($Y = 0$).



(a) (b)

Fig. 6-25 (a) Wired-AND logic is obtained by connecting the outputs of positive NAND gates together; (b) a two-input and a three-input NAND gate wired-AND together to perform the logic in Eq. (6-36).

Consider two positive NAND gates wired-AND together as in Fig. 6-25*b*. Then $Y_1 = \overline{AB}$ and $Y_2 = \overline{CDE}$. Hence

$$Y = Y_1Y_2 = (\overline{AB})(\overline{CDE}) = \overline{AB + CDE} \qquad (6\text{-}36)$$

where use is made of De Morgan's law. Note that the wired-AND has led to an implementation of the AOI two-level logic (Sec. 6-7). Because of the + sign in Eq. (6-36), the connection in Fig. 6-25 is often incorrectly referred to as a positive wired-OR.

Note that the wired-AND connection places the collector resistors in Fig. 6-25*a* in parallel. This reduction in resistance increases the power dissipation in the ON state. In order to avoid this condition *open-collector* gates are available specifically for wired-AND applications.

6-11 HIGH-THRESHOLD-LOGIC (HTL) GATE[5]

In an industrial environment the noise level is quite high because of the presence of motors, high-voltage switches, on-off control circuits, etc. By using a higher supply voltage (15 V instead of 5 V) and a 6.9-V Zener diode in place of *D*2 in the DTL gate of Fig. 6-24, this circuit is converted into the high-noise-immunity gate of Fig. 6-26. The resistances are increased in Fig. 6-26 with respect to those in Fig. 6-24, so that approximately the same currents are obtained in both circuits. The noise margin obtained with this circuit is typically 7 V (Prob. 6-44).

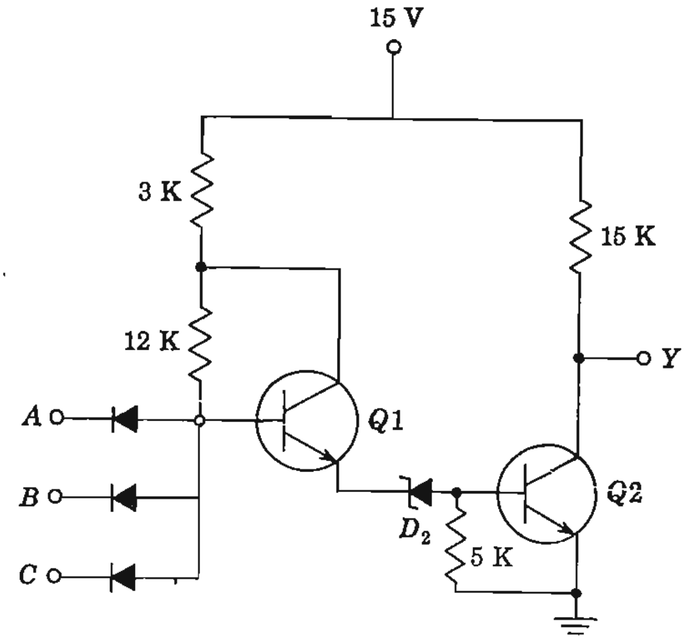


Fig. 6-26 A high-threshold-logic NAND gate.

6-12 TRANSISTOR–TRANSISTOR–LOGIC (TTL) GATE[6]

The fastest-saturating logic circuit is the transistor-transistor-logic gate (TTL, or T^2L), shown in Fig. 6-27. This switch uses a multiple-emitter transistor which is easily and economically fabricated using integrated-circuit techniques (Sec. 7-7). The TTL circuit has the topology of the DTL circuit of Fig. 6-23, with the emitter junctions of $Q1$ acting as the input diodes D of the DTL gate and the collector junction of $Q1$ replacing the diode $D1$ of Fig. 6-23. The base-to-emitter diode of $Q2$ is used in place of the diode $D2$ of the DTL gate, and both circuits have an output transistor ($Q3$ or Q).

The explanation of the operation of the TTL gate parallels that of the DTL switch. Thus, if at least one input is at $V(0) = 0.2$ V, then

$$V_P = 0.2 + 0.7 = 0.9\text{V}$$

For the collector junction of $Q1$ to be forward-biased and for $Q2$ and $Q3$ to be ON requires about $0.7 + 0.7 + 0.7 = 2.1$ V. Hence these are OFF; the output rises to $V_{CC} = 5$ V, and $Y = V(1)$. On the other hand, if all inputs are high (at 5 V), the input diodes (the emitter junctions) are reverse-biased and V_P rises toward V_{CC} and drives $Q2$ and $Q3$ into saturation. Then the output is $V_{CE,\text{sat}} = 0.2$ V, and $Y = V(0)$ (and V_P is clamped at about 2.3 V).

The above explanation indicates that $Q1$ acts like isolated back-to-back diodes, and not as a transistor. However, we shall now show that, during turnoff, $Q1$ does exhibit transistor action, thereby reducing storage time (Sec. 6-5) considerably. Note that the base voltage of $Q2$, which equals the collector voltage of $Q1$, is at $0.8 + 0.8 = 1.6$ V during saturation of $Q2$ and $Q3$. If now any input drops to 0.2 V, then $V_P = 0.9$ V, and hence the base of $Q1$ is at 0.9 V. At this time the collector junction is reverse-biased by

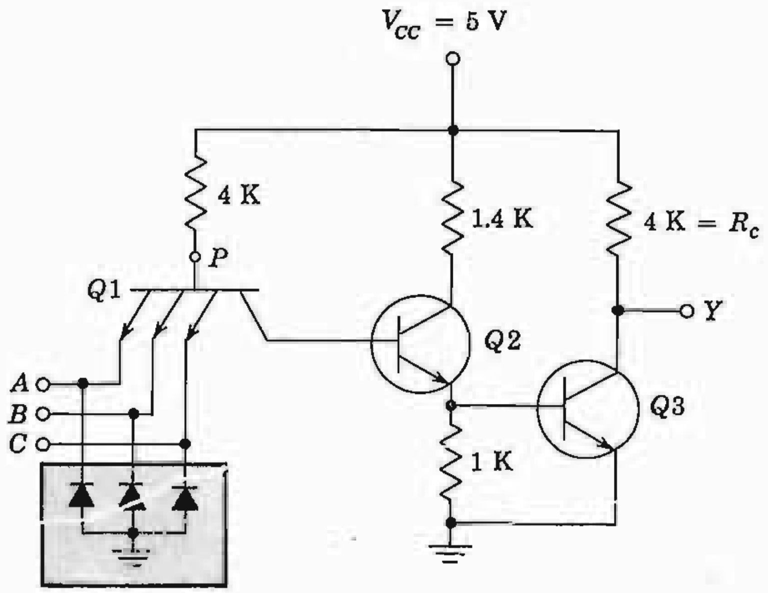


Fig. 6-27 An IC positive TTL NAND gate. (Neglect the diodes in the shaded block.)

$1.6 - 0.9 = 0.7$ V, the emitter junction is forward-biased, and $Q1$ is in its active region. The large collector current of $Q1$ now quickly removes the stored charge in $Q2$ and $Q3$. It is this transistor action which gives TTL the highest speed of any saturated logic.

Clamping diodes (shown in the shaded block in Fig. 6-27) are often included from each input to ground, with the anode grounded. These diodes are effectively out of the circuit for positive input signals, but they limit negative voltage excursions at the input to a safe value. These negative signals may arise from ringing caused by lead inductance resonating with shunt capacitance.

6-13 OUTPUT STAGES

At the output terminal of the DTL or TTL gate there is a capacitive load C_L, consisting of the capacitances of the reverse-biased diodes of the fan-out gates and any stray wiring capacitance. If the collector-circuit resistor of the inverter is R_c (called a *passive pull-up*), then, when the output changes from the low to the high state, the output transistor is cut off and the capacitance charges exponentially from $V_{CE,\text{sat}}$ to V_{CC}. The time constant R_cC_L of this waveform may introduce a prohibitively long delay time into the operation of these gates.

The output delay may be reduced by decreasing R_c, but this will increase the power dissipation when the output is in its low state and the voltage across R_c is $V_{CC} - V_{CE,\text{sat}}$. A better solution to this problem is indicated in Fig. 6-28, where a transistor acts as an *active pull-up* circuit, replacing the passive pull-up resistance R_c. This output configuration is called a *totem-pole* amplifier because the transistor $Q4$ "sits" upon $Q3$. It is also referred to as a power-driver, or power-buffer, output stage.

The transistor $Q2$ acts as a *phase splitter*, since the emitter voltage is out of phase with the collector voltage (for an increase in base current, the emitter voltage increases and the collector voltage decreases). We now explain the operation of this driver circuit in detail, with reference to the TTL gate of Fig. 6-28*a*.

The output is in the low-voltage state when $Q2$ and $Q3$ are driven into saturation. For this state we should like $Q4$ to be OFF. Is it? Note that the collector voltage V_{CN2} of $Q2$ with respect to ground N is given by

$$V_{CN2} = V_{CE2,\text{sat}} + V_{BE3,\text{sat}} = 0.2 + 0.8 = 1.0 \text{ V} \tag{6-37}$$

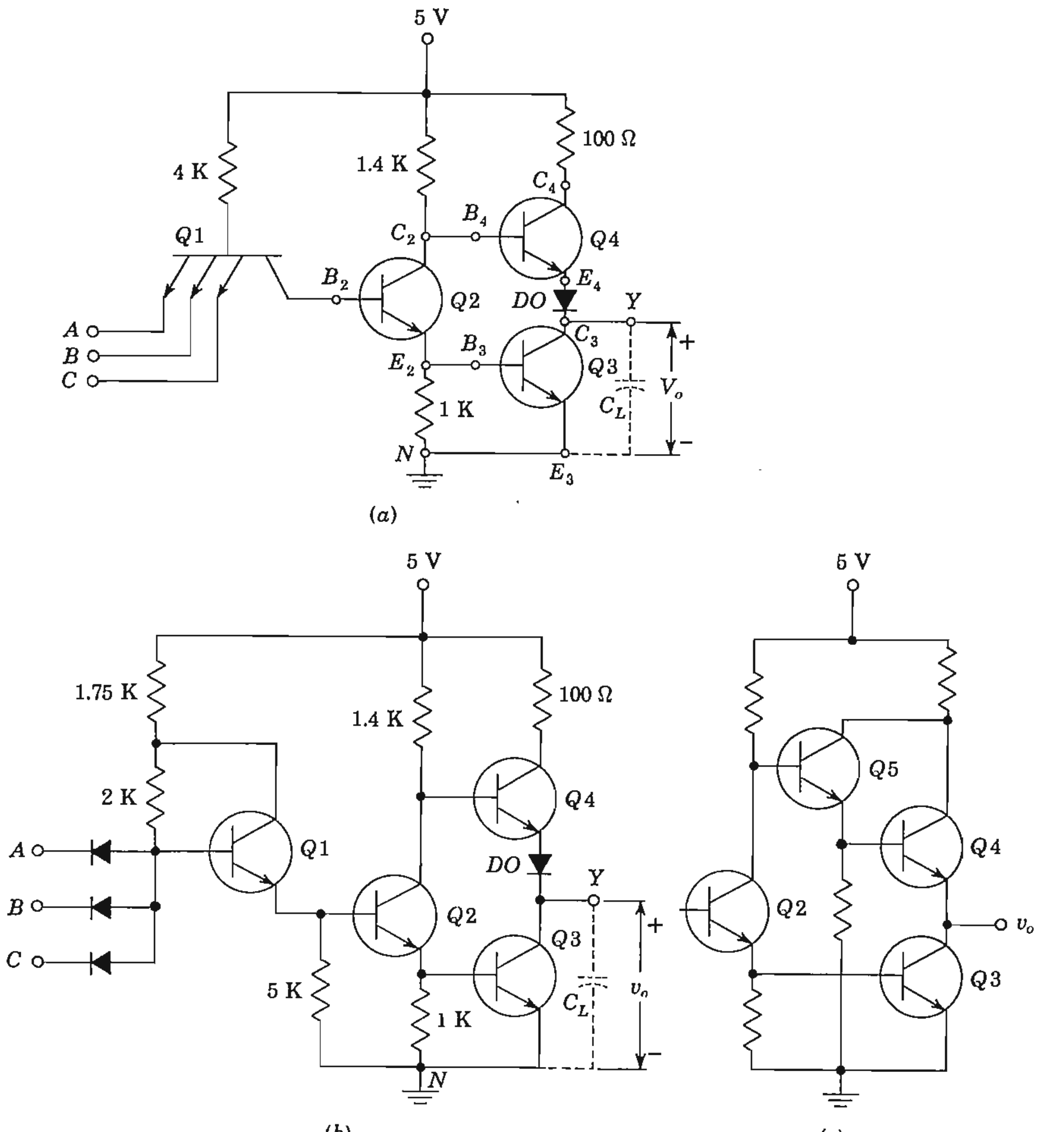


Fig. 6-28 (*a*) A TTL gate and (*b*) a DTL gate with a totem-pole output driver. (*c*) A modified configuration of the output circuit. [The circuit to the left of $Q2$ is identical with that in (*a*) for TTL or in (*b*) for DTL.]

Since the base of $Q4$ is tied to the collector of $Q2$, then $V_{BN4} = V_{CN2} = 1.0$ V. *If the output diode DO were missing,* the base-to-emitter voltage of $Q4$ would be

$$V_{BE4} = V_{BN4} - V_{CE3,\text{sat}} = 1.0 - 0.2 = 0.8 \text{ V}$$

which would put $Q4$ into saturation. Under these circumstances the steady current through $Q4$ would be

$$\frac{V_{CC} - V_{CE4,\text{sat}} - V_{CE3,\text{sat}}}{100} = \frac{5 - 0.2 - 0.2}{100} \text{ A} = 46 \text{ mA} \tag{6-38}$$

which is excessive and wasted current. The necessity for adding DO is now clear. With it in place, the sum of V_{BE4} and V_{DO} is 0.8 V. Hence both $Q4$ and DO are at cutoff. In summary, if C_L is at the high voltage $V(1)$ and the gate is excited, $Q4$ and DO go OFF, and $Q3$ conducts. Because of its large active-region current, $Q3$ quickly discharges C_L, and as v_o approaches $V(0)$, $Q3$ enters saturation. The bottom transistor $Q3$ of the totem pole is referred to as a *current sink,* which discharges C_L.

Assume now that with the output at $V(0)$, there is a change of state, because one of the inputs drops to its low state. Then $Q2$ is turned OFF, which causes $Q3$ to go to cutoff because V_{BE3} drops to zero. The output v_o remains momentarily at 0.2 V because the voltage across C_L cannot change instantaneously. Now $Q4$ goes into saturation and DO conducts, as we can verify:

$$V_{BN4} = V_{BE4,\text{sat}} + V_{DO} + v_o = 0.8 + 0.7 + 0.2 = 1.7 \text{ V}$$

and the base and collector currents of $Q4$ are

$$I_{B4} = \frac{V_{CC} - V_{BN4}}{1.4} = \frac{5 - 1.7}{1.4} = 2.36 \text{ mA}$$

$$I_{C4} = \frac{V_{CC} - V_{CE4,\text{sat}} - V_{DO} - v_o}{0.1} = \frac{5 - 0.2 - 0.7 - 0.2}{0.1} = 39.0 \text{ mA}$$

Hence, if h_{FE} exceeds $(h_{FE})_{\min} = I_{C4}/I_{B4} = 39.0/2.36 = 16.5$, then $Q4$ is in saturation. The transistor $Q4$ is referred to as a *source,* supplying current to C_L. As long as $Q4$ remains in saturation, the output voltage rises exponentially toward V_{CC} with the very small time constant $(100 + R_{CS4} + R_f)C_L$, where R_{CS4} is the saturation resistance (Sec. 5-8) of $Q4$, and where R_f (a few ohms) is the diode forward resistance. As v_o increases, the currents in $Q4$ decrease, and $Q4$ comes out of saturation and finally v_o reaches a steady state when $Q4$ is at the cutin condition. Hence the final value of the output voltage is

$$v_o = V_{CC} - V_{BE4,\text{cutin}} - V_{DO,\text{cutin}} \approx 5 - 0.5 - 0.6 = 3.9 \text{ V} = V(1) \tag{6-39}$$

If the 100-Ω resistor were omitted; there would result a faster change in output from $V(0)$ to $V(1)$. However, the 100-Ω resistor is needed to limit the current spikes during the turn-on and turn-off transients. In particular, $Q3$ does not turn off (because of storage time) as quickly as $Q5$ turns on. With both totem-pole transistors conducting at the same time, the supply voltage would be short-circuited if the 100-Ω resistor were missing. The peak current drawn from the supply during the transient is limited to $I_{C4} + I_{B4} = 39 + 2.4 \approx 41$ mA if the 100-Ω resistor is used. These current spikes generate noise in the power-supply distribution system, and also result in increased power consumption at high frequencies.

Wired Logic It should be emphasized that the wired-AND connection must *not* be used with the totem-pole driver circuit. If the output from one gate is high while that from a second gate is low, and if these two outputs are tied together, we have exactly the situation just discussed in connection with transient current spikes. Hence, if the wired-AND were used, the power supply would deliver a *steady* current of 41 mA under these circumstances.

Alternative Output Stages From the foregoing discussion it should be clear that the diode *DO* can be moved from the emitter into the base lead of *Q*4. This configuration is used by some manufacturers.

The diode in the base of *Q*4 referred to in the preceding paragraph may be the base-to-emitter diode of an additional transistor, such as *Q*5 in Fig. 6-28*c*. The combination of *Q*5 and *Q*4, called a *Darlington pair*, is discussed in Sec. 8-6. It has a very high current gain and extremely low output resistance and results in a very high speed gate. Another form of power buffer is given in Prob. 6-45.

Storage time is eliminated by using *Schottky transistors* (which are clamped to prevent saturation, Sec. 7-12) for the transistors in Fig. 6-28.

6-14 RESISTOR–TRANSISTOR LOGIC (RTL) AND DIRECT–COUPLED TRANSISTOR LOGIC (DCTL)

In 1972 TTL was the most popular family, and it had captured about 45 percent of the integrated-circuit digital market. Approximately another 30 percent of the market was taken by DTL, and the remaining 25 percent was divided between several additional families, called *resistor-transistor logic* (RTL), *direct-coupled transistor logic* (DCTL), *emitter-coupled logic* (ECL), and *metal-oxide-semiconductor* (MOS) *logic*. Since MOS logic uses the field-effect transistor (FET), the discussion of these gates is postponed until Chap. 10, where the MOSFET is introduced. Since ECL requires an understanding of the *differential amplifier*, this type of (nonsaturating) logic is considered in Sec. 16-17, after the differential amplifier is studied. A discussion of RTL and DCTL now follows.

Resistor-Transistor Logic (RTL)[5] This configuration is indicated in Fig. 6-29*a*, which represents a three-input positive NOR gate with a fan-out of 5. If any input is high, the corresponding transistor is driven into saturation and the output is low, $v_o = V_{CE,sat} \approx 0.2$ V. However, if all inputs are low, then all input transistors are cut off by $V_\gamma - V(0) = 0.5 - 0.2 = 0.3$ V and the

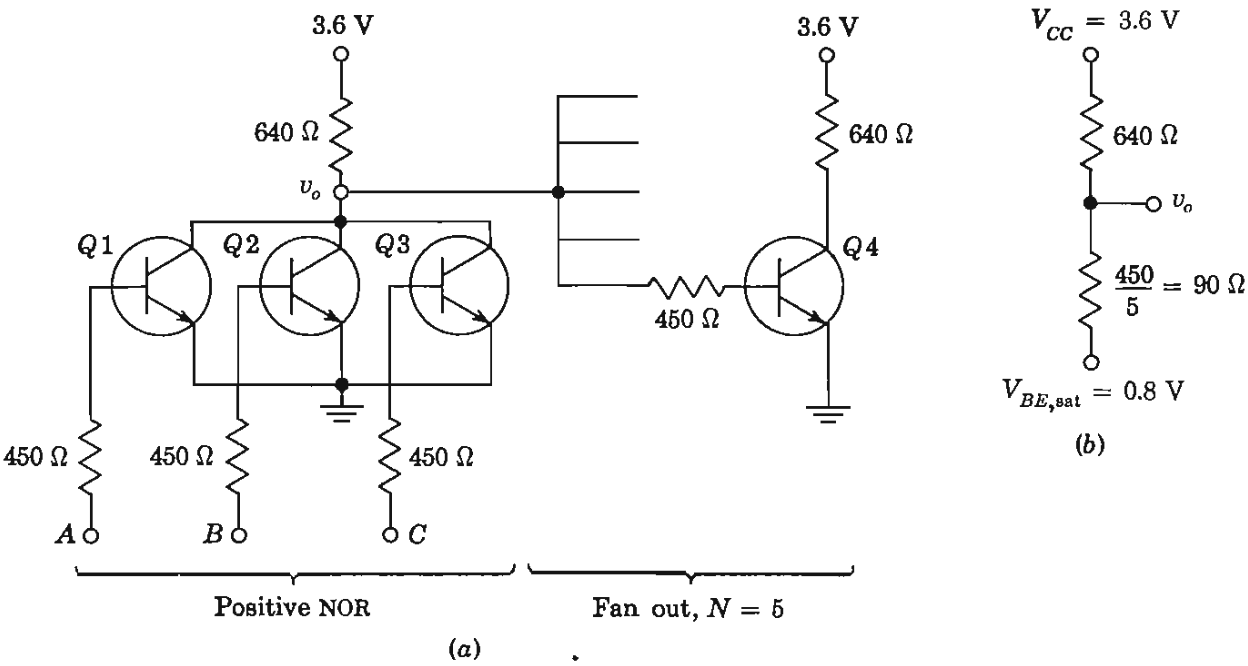


Fig. 6-29 (a) An RTL positive NOR gate with a fan-in of 3 and a fan-out of 5. (b) The equivalent circuit from which to calculate v_o in the high state.

output v_o is high. (Note the low noise margin.) The preceding two statements confirm that the gate performs positive NOR (or negative NAND) logic.

The value of v_o depends upon the fan-out. For example, if $N = 5$, then the output of the NOR gate is loaded by five 450-Ω resistors in parallel (or 90 Ω), which is tied to $V_{BE,\text{sat}} \approx 0.8$ V, as shown in Fig. 6-29b. Under these circumstances (using superposition),

$$v_o = \frac{3.6 \times 90}{90 + 640} + \frac{0.8 \times 640}{90 + 640} = 1.14 \text{ V} \tag{6-40}$$

This voltage must be large enough so that the base current can drive each of the five transistors into saturation. Since

$$I_B = \frac{1.14 - 0.8}{0.45} = 0.755 \text{ mA} \qquad I_C = \frac{3.6 - 0.2}{0.64} = 5.31 \text{ mA}$$

then the circuit will operate properly if $h_{FE} > (h_{FE})_{\min} = 5.31/0.76 = 7.0$.

Resistor-transistor logic uses the minimum space (for a standard digital function) on a silicon wafer, and hence is very economical.

Direct-Coupled Transistor Logic (DCTL)[7] This configuration is the same as RTL, except that the base resistors are omitted. In Fig. 6-30 the fan-in is 3 and the fan-out is 2.

To verify that the circuit implements positive NOR logic, consider first that all inputs are in the 0 state. Because this low voltage to an input (say, to $Q1$) comes from a saturated transistor (Q') of a preceding state,

$$v_1 = V_{CE,\text{sat}} = V(0)$$

Since this voltage is 0.2 V for a saturated silicon transistor, and since the cutin voltage $V_\gamma \approx 0.5$ V, $Q1$ will conduct very little (although the noise margin is only $0.5 - 0.2 = 0.3$ V as it is with RTL logic). Since the current in $Q1$ is almost zero, the output Y tries to raise V_{CC} and $Q4$ and $Q5$ go into saturation. Hence the output Y is clamped at

$$V_{BE,\text{sat}} = V(1) \approx 0.8 \text{ V}$$

for silicon. Thus, with all inputs in the low state, the output is in the high state. Note that the high state is only 0.8 V, independent of V_{CC}.

Consider now that at least one input v_1 is in the high state. Since $Q1$ is fed from Q', Q' is cut off and $Q1$ is driven into saturation. Under these circumstances the output Y is $V_{CE,\text{sat}} = V(0)$. If more than one input is excited, the output will certainly be low. Hence we have confirmed that the NOR function is satisfied.

There are a number of difficulties with DCTL: (1) The reverse saturation current for all fan-in transistors adds in the common collector-circuit resistor R_c. At high enough temperatures the total $I_{CBO}R_c$ drop may be large enough so that the output Y is too low to drive the fan-out transistors into saturation. (2) Because of the direct connection, the base current is almost equal to the collector current (for $V_{CC} \gg V_{CE,\text{sat}}$ and $V_{CC} \gg V_{BE,\text{sat}}$). With a transistor so heavily driven into saturation, very large stored base charge will result, with a corresponding detrimental effect on the switching speed. (3) Since the voltage levels are so low—the total output-voltage step is only of the order of 0.6 V for silicon—then spurious (noise) spikes can be troublesome. (4) The bases of the fan-out transistors are connected together. Since the input characteristics can never be identical, let us assume that $Q4$ has a much lower V_{BE} for a given I_B than does $Q5$. Under these circumstances, $Q4$ will "hog" most of the base current, and it is possible that $Q5$ may not even be driven into saturation. Hence transistors suitable for DCTL must have very

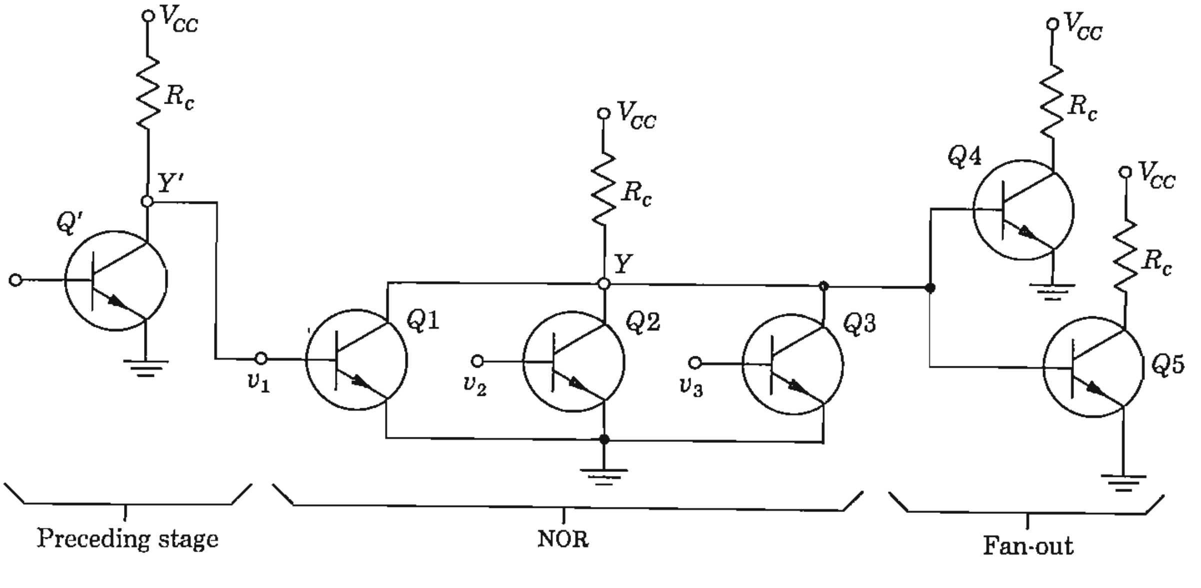


Fig. 6-30 A positive NOR DCTL gate.

close control on uniformity of input characteristics, very low values of I_{CBO}, as large a differential as possible between $V_{BE,\text{sat}}$ and $V_{CE,\text{sat}}$, a large h_{FE}, and a small storage time.

The advantages of DCTL are: (1) Only one low-voltage supply (operation with 1.5 V is possible) is needed; (2) transistors with low breakdown voltages may be used; and (3) the power dissipation is low.

In DCTL a NOR or NAND circuit is possible in which the transistors are in series (totem-pole fashion) rather than in parallel as in Fig. 6-30. Because of the difficulties mentioned above, DCTL is not often used with bipolar transistors, but it is the standard logic with field-effect transistors (Sec. 10-6).

6-15 COMPARISON OF LOGIC FAMILIES

In the discussion of each logic configuration some of its advantages and disadvantages have been listed. An exhaustive comparison is extremely difficult because we must take into account all the following characteristics: (1) speed (propagation time delay), (2) noise immunity, (3) fan-in and fan-out capabilities, (4) power-supply requirements, (5) power dissipation per gate, (6) suitability for integrated fabrication, (7) operating temperature range, (8) number of functions available, and (9) cost. Also to be considered is the personal prejudice of the engineer, who is always strongly influenced by past experience. Items 1 and 8 require some explanation; all others have already been defined or are self-evident.

Propagation Delay In Fig. 6-9 the ON time t_{ON} of an (n-p-n) inverter is defined as the interval between the application of an ideal pulse input and the time when the output current reaches 90 percent (or the output voltage falls to 10 percent) of its final value. This definition is *not* useful for the propagation-delay time of a logic gate, for two reasons. First, the input to a gate is not a square pulse, but it is a waveform with a nonzero rise time. Second, the input does not have to reach its 90 percent value before the gate changes state. Hence we define *propagation-delay* ON *time* T_{ON} (to distinguish it from t_{ON}) as indicated below.

As the input voltage to a positive NAND gate rises from $V(0)$ toward $V(1)$, then at some *switching threshold voltage* $V'(0)$ (Fig. 6-31), conditions within the gate are modified, so that a change of state of the output from $V(1)$ to $V(0)$ is initiated. Similarly, as the input falls from $V(1)$ toward $V(0)$, then at some other *switching threshold voltage* $V'(1)$, the initiation of the change of state from $V(0)$ to $V(1)$ takes place. We now define (as in Fig. 6-31) the propagation delay ON time T_{ON} as the interval between the time when the input v_i reaches $V'(0)$ and the output falls to $V'(1)$. Also, the propagation delay OFF time T_{OFF} is defined as the interval between the time when the input equals $V'(1)$ and the output rises to $V'(0)$. Because of minority-

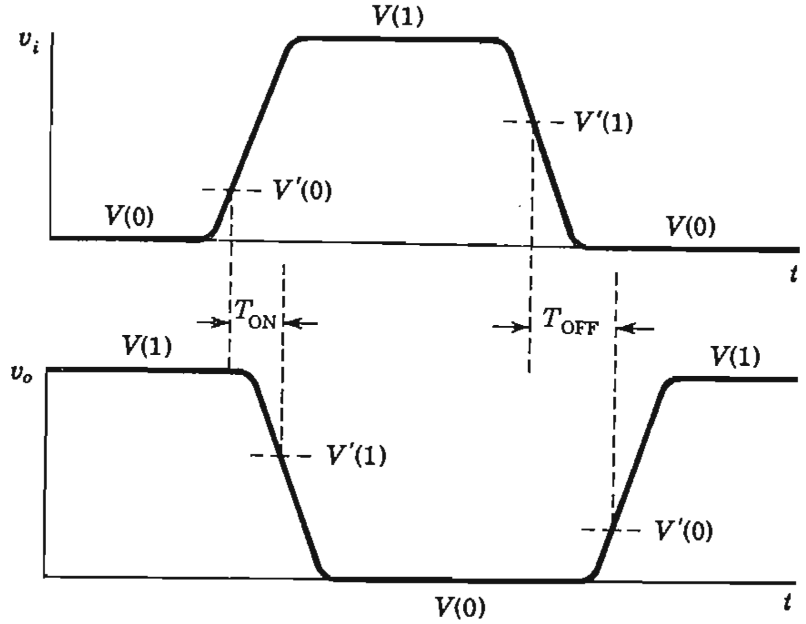


Fig. 6-31 Pertaining to the definitions of the propagation-delay times.

carrier storage time, $T_{ON} \neq T_{OFF}$. Hence the *propagation-delay time* T_{PD} is usually defined as the average of these two times, or

$$T_{PD} = \tfrac{1}{2}(T_{ON} + T_{OFF}) \tag{6-41}$$

In passing, we note that some authors arbitrarily assume the two threshold voltages to be equal: $V'(0) = V'(1) = \frac{1}{2}[V(0) + V(1)]$.

Functions The basic AND, OR, NAND, and NOR gates are combined in one integrated chip in various combinations to perform specific circuit func-

TABLE 6-5 Comparison of the major IC digital logic families[8]

Parameter \ Logic	RTL	DTL	HTL	TTL	ECL	MOS	CMOS
Basic gate[a]	NOR	NAND	NAND	NAND	OR-NOR	NAND	NOR or NAND
Fan-out[b]	5	8	10	10	25	20	>50
Power dissipated[c] per gate, mW	12	8–12	55	12–22	40–55	0.2–10	0.01 static 1 at 1 MHz
Noise immunity	Nominal	Good	Excellent	Very good	Good	Nominal	Very good
Propagation delay[d] per gate, ns	12	30	90	12–6	4–1	300	70
Clock rate,[e] MHz	8	12–30	4	15–60	60–400	2	5
Number of functions[f]	High	Fairly high	Nominal	Very high	High	Low	Low
Reference	Sec. 6-14	Sec. 6-10	Sec. 6-11	Sec. 6-12	Sec. 16-17	Sec. 10-7	Sec. 10-7

[a] Positive logic.
[b] Worst-case number of inputs that the gate can drive.
[c] Typical; affected by temperature and frequency of operation.
[d] Typical for a nominal fan-out.
[e] Maximum frequency at which flip-flops operate. The actual clock rate is from one-half to one-tenth the frequency listed.
[f] New functions are being added continuously by the manufacturers, except to DTL. However, TTL functions are compatible with DTL.

tions. These *system building blocks* are discussed in Chap. 17 and include flip-flops, counters, arithmetic functions, decoders, shift registers, etc.

It should be clear that no single logic configuration can be best suited for all applications. A comparison of the major integrated-circuit digital logic families is given in Table 6-5.

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REVIEW QUESTIONS

6-1 Express the following decimal numbers in binary form: (*a*) 28; (*b*) 100; (*c*) 5,127.

6-2 Define (*a*) *positive logic;* (*b*) *negative logic.*

6-3 Define an OR gate and give its truth table.

6-4 Draw a positive diode OR gate and explain its operation.

6-5 Evaluate the following expressions: (*a*) $A + 1$; (*b*) $A + A$; (*c*) $A + 0$.

6-6 Define an AND gate and give its truth table.

6-7 Draw a positive-diode AND gate and explain its operation.

6-8 Evaluate the following expressions: (*a*) $A1$; (*b*) AA; (*c*) $A0$; (*d*) $A + AB$.

6-9 Define a NOT gate and give its truth table.

6-10 Draw a positive-logic NOT gate and explain its operation.

6-11 Evaluate the following expressions: (*a*) $\bar{\bar{A}}$; (*b*) $\bar{A}A$; (*c*) $\bar{A} + A$.

6-12 A pulse waveform drives an *n-p-n* transistor from cutoff into saturation and then back to cutoff. (*a*) Draw the output current waveshape, lined up in time with the input voltage. (*b*) Indicate the following times on your sketch: *delay, rise,* ON, *storage, fall,* and OFF.

6-13 (*a*) What is the physical origin of *storage time?* (*b*) Is it important in turning a transistor ON or OFF? Explain. (*c*) Draw the minority-carrier concentration in the base; in the active region and in saturation.

6-14 Define an INHIBITOR and give the truth table for $AB\bar{C}$.

6-15 Define an EXCLUSIVE OR and give its truth table.

6-16 Show two logic block diagrams for an EXCLUSIVE OR.

6-17 Verify that the following Boolean expressions represent an EXCLUSIVE OR: (*a*) $\overline{AB + \bar{A}\bar{B}}$; (*b*) $(A + B)(\bar{A} + \bar{B})$.

6-18 State the two forms of De Morgan's laws.

6-19 Show how to implement an AND with OR and NOT gates.

6-20 Show how to implement an OR with AND and NOT gates.

6-21 Define a NAND gate and give its truth table.

6-22 Draw a positive NAND gate with diodes and a transistor (DTL) and explain its operation.

6-23 Define a NOR gate and give its truth table.

6-24 Repeat Rev. 6-22 for a NOR gate.

6-25 Draw the circuit of an IC DTL gate and explain its operation.

6-26 Define (*a*) *fan-out;* (*b*) *fan-in;* (*c*) *standard load;* (*d*) *current sink;* (*e*) *current source.*

6-27 What logic is performed if the outputs of two DTL gates are connected together? Explain.

6-28 How does high-threshold logic (HTL) differ from DTL?

6-29 Draw the circuit of a TTL gate and explain its operation.

6-30 Draw a totem-pole output buffer with a TTL gate. Explain its operation.

6-31 Repeat Rev. 6-30 for a DTL gate.

6-32 Draw the circuit of an RTL gate and explain its operation for positive logic.

6-33 Repeat Rev. 6-32 for negative logic.

6-34 Draw a DCTL circuit and explain its operation.

6-35 List three advantages and three disadvantages of DCTL gates.

6-36 Define (*a*) two *threshold voltages;* (*b*) *propagation-delay* ON *time;* (*c*) *propagation-delay* OFF *time;* (*d*) *propagation-delay time.*

6-37 Give an order of magnitude which is applicable to any of the logic families studied in this chapter for (*a*) fan-out; (*b*) power dissipation per gate; (*c*) propagation delay per gate; (*d*) clock rate.